



AKTU

B.Tech III-Sem / MCA



CS IT & CS Allied

COA Computer Organization & Architecture

Crash Course



Unit-1 in One Shot



By Pragya Ma'am

Syllabus Unit-I

Introduction: Functional units of digital system and their interconnections, buses, bus architecture, types of buses and bus arbitration. Register, bus and memory transfer. Processor organization, general registers organization, stack organization and addressing modes.

Most Important Questions: Unit-1

Q.1	List and briefly define the main structural component of a computer. Or Explain functional unit of computer system in detail.
Q.2	What is bus? what is system bus? What is bus arbitration and why is need? also explain the type of bus architecture
Q.3	How memory read and write operation perform in computer system?
Q.4	What is bus transfer and memory transfer? How bus transfer happen through multiplexer and three state buffer?
Q.5	What is addressing mode? Explain different type of addressing modes.
Q.6	What is processor organisation? And explain- 1 – Accumulator based 2 – Stag organisation 3 – General register

Computer Organization and Architecture

UNIT 1 INTRODUCTION

Today's Target

- Functional Unit of Digital System
- Buses
- AKTU PYQs

By PRAGYA RAJVANSHI

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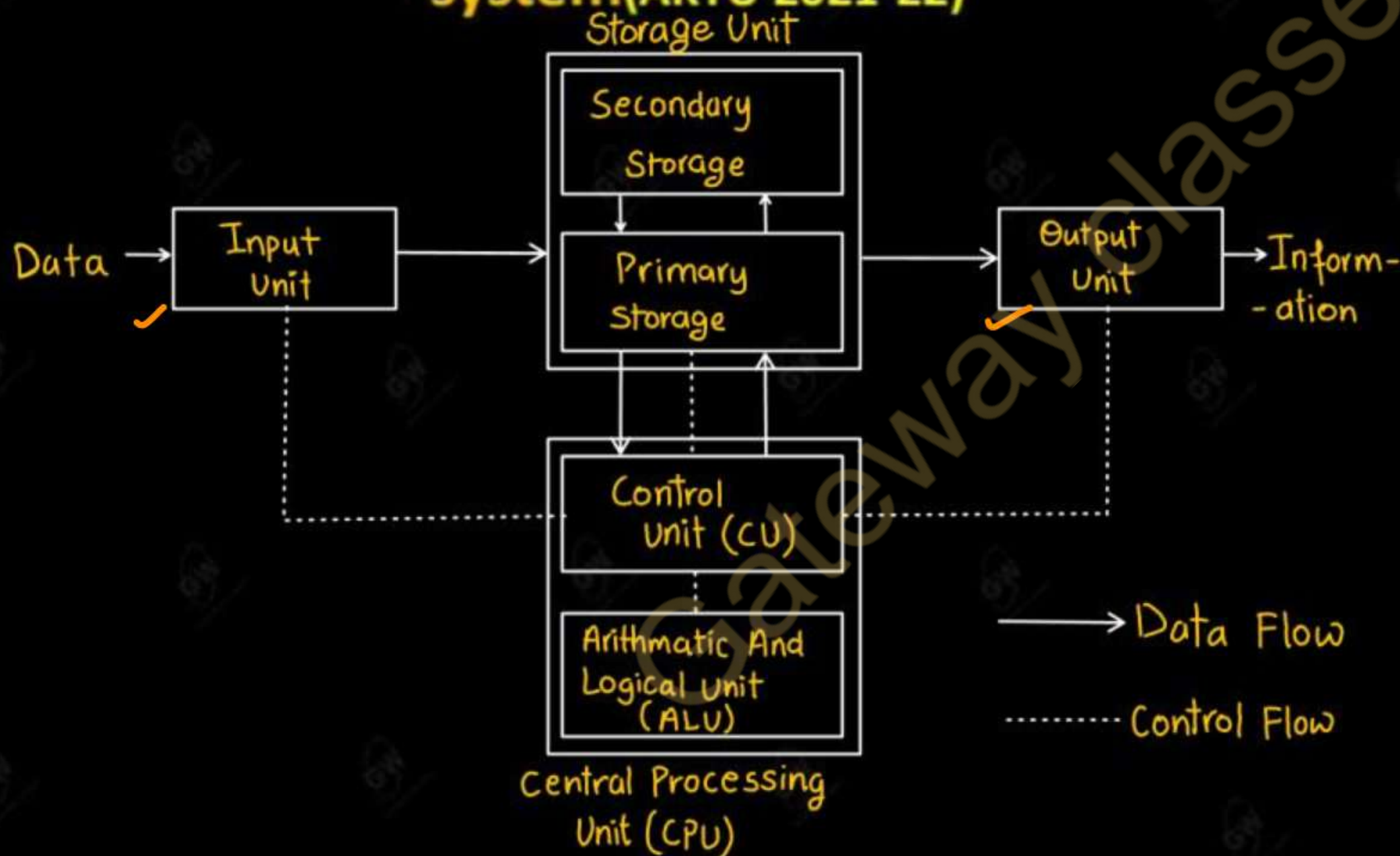
Difference between (AKTU 2020-21)

Computer Architecture	Computer Organization
Architecture describes what the computer does.	The Organization describes how it does it.
It deals with high-level design issues.	It deals with low-level design issues.
Computer Architecture deals with the <u>functional</u> behavior of computer systems.	Computer Organization deals with a <u>structural</u> <u>relationship</u> .
✓ Architecture indicates its hardware.	Organization indicates its <u>performance</u> .

Functional Unit of Digital System

(AKTU 2019-20/2022-23)

Main Structural Component of the Computer system (AKTU 2021-22)



Input Unit:

- We need to first enter the data & instruction in the computer system, before any computation begins.
- This task is done by the input devices. (keyboard, mouse, scanner, digital camera).

(5 Structural Components)

- Storage → Input
- ALU → Output
- CU

Output Unit:

- Output unit accepts the results produced by the computer in coded form. (low level lang)
- It converts these coded results to human readable form.
- It displays the converted results to the outside world with the help of output devices (monitors, printers, projectors).

Storage Unit:

- This storage unit is designed to save the initial data, the intermediate result & the final result.

1.Primary Storage(main memory):

- It is computer memory that a processor or computer accesses first or directly.
- Faster access speed.
- It is of two types RAM and ROM.
- Smaller storage capacity.

1.1. RAM (Random Access memory)

- Volatile (loses data when power is off).
- It is also known as read/write memory.
- Used for temporary data storage and quick access by the CPU.

RAM are of two type static RAM (SRAM) and DRAM (Dynamic RAM). (for few milliseconds)

1.2. ROM (Read only memory)

- Non-Volatile (retains data when power is off).
- ROM provides the necessary instructions to load the operating system from secondary storage into RAM during the boot process.
- ROM holds data that doesn't need to be modified, such as system-level instructions, bootstrapping data, or default settings.

2. Secondary Storage (Auxiliary storage)

- It can retain information even when the system is off.
- It is basically used for holding the program instructions & data on which the computer is not working on currently, but needs to process them later.

✓ PROM (write once read many)
✓ EPROM (UV light)
✓ EEPROM (Electrical waves)

Central Processing Unit:

- It has two major components ALU and CU
- The CPU is the brain of the computer.

1. Arithmetic Logic Unit:

- The actual execution of the instructions (arithmetic or logical operations) takes place over here.
*+ , - * / % . log .*
- The ALU performs simple addition, subtraction, multiplication, division, and logic operations, such as OR and AND.

2 Control Unit:

- This unit controls the operations of all parts of the computer but does not carry out any actual data processing.
- It manages and coordinates all the units of the system.

(Another way this question may be asked.

Explain the arrangement of CPU With
memory, i/o)

What is bus? (AKTU 2017-18)

- A bus is a communication system that transfers **data** between components
- It consists of a set of parallel wires or lines that carry data, control signals, and addresses *(read, write)*
- It enabling various parts of a computer, like the CPU, memory, and peripherals, to **communicate with each other efficiently.**

What is System Bus?

- A bus that connects major components (CPU, memory and I/O devices) of a computer system
- It includes the data bus, address bus, and control bus, which work together to manage **data flow and communication** throughout the system.

✓ Why do we need bus?

Communication:

- Buses facilitate communication between the **CPU, memory, and peripheral devices**, allowing data and instructions to be transferred efficiently.

Coordination:

- They manage and **coordinate the flow of data and control signals**, ensuring that different components of the computer work together harmoniously.

Scalability:

- Buses allow for the **expansion of computer systems** by connecting additional components and peripherals, such as expansion cards or external devices.

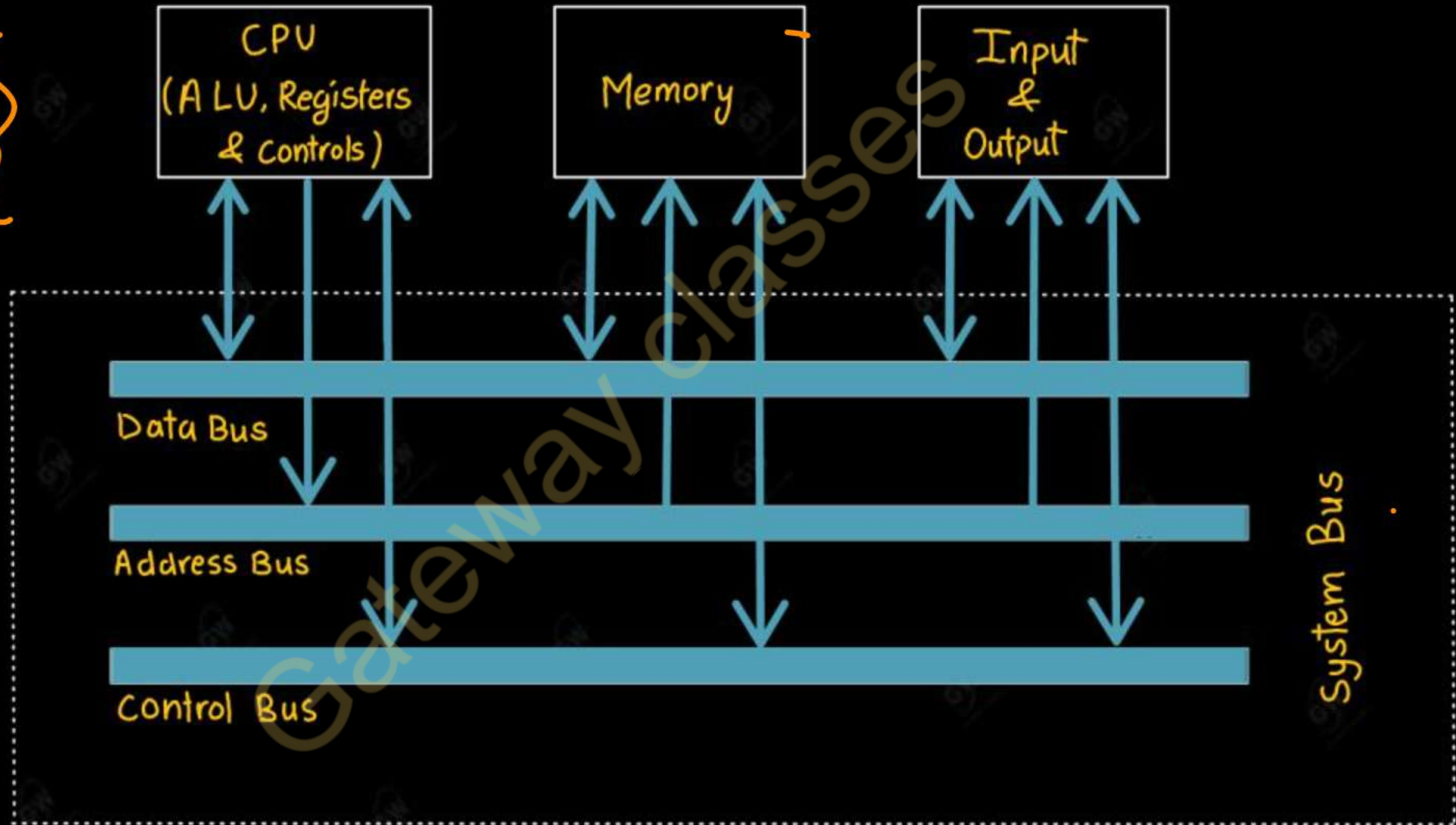
Efficiency:

- By using shared pathways (buses), multiple components **can access the system's resources** without needing **individual connections for each device**, which simplifies the design and reduces costs.

MAJOR COMPONENT OF SYSTEM BUS (AKTU 2015-16)

- Data Bus
- Address Bus
- Control Bus

types
of Buses



DATA BUS

CONTROL BUS (AKTU 2017-18/18-19)

ADDRESS BUS

It is used for transmitting the data / instruction from CPU to memory/IO and vice-versa.

It is used to transfer the control and timing signals from one component to the other component.

It is used to carry address from CPU to memory/IO devices.

It is bi-directional.

It is bi-directional.

It is uni-directional.

NOTE: Control signals are generated in the control unit of CPU.

Memory read – Data from memory address location to be placed on data bus.

Memory write – Data from data bus to be placed on memory address location.

I/O Read – Data from I/O address location to be placed on data bus.

I/O Write – Data from data bus to be placed on I/O address location.

Timing signals are used to synchronize the memory and I/O operations with a CPU clock.

Simplified communication:

- It provides a standardized way for different components (CPU, memory, peripherals) to communicate, reducing complexity.

Cost-effective:

- Using a common bus for data transfer reduces the need for multiple separate connections, saving on hardware costs.

Scalability:

- System buses can be easily expanded to support additional devices, making system upgrades simpler.

Bottleneck risk:

- All data transfers share the same bus, which can lead to congestion and slow performance as the number of devices increases.

Limited bandwidth:

- The bus has a finite bandwidth, restricting the speed at which data can be transferred between components.

Shared resources:

- Devices on the bus compete for the same resources, which can lead to delays and inefficiency in data processing.

AKTU PYQs

Q.1 ✓	What do you mean by bus? types of Buses → Address → Control → Data	AKTU 2018-19
Q.2 ✓ <u>Sub</u>	List and briefly define the main structural components of a computer OR Explain functional unit of computer system in detail.	AKTU 2021-22 AKTU 2022-23
Q.3 ✓	Define the term computer architecture and computer organization.	AKTU 2020-21

Computer organization and architecture

UNIT 1 INTRODUCTION

Today's Target

- Types of buses
- Register and its types
- Idea about bus arbitration
- AKTU PYQs

By PRAGYA RAJVANSHI

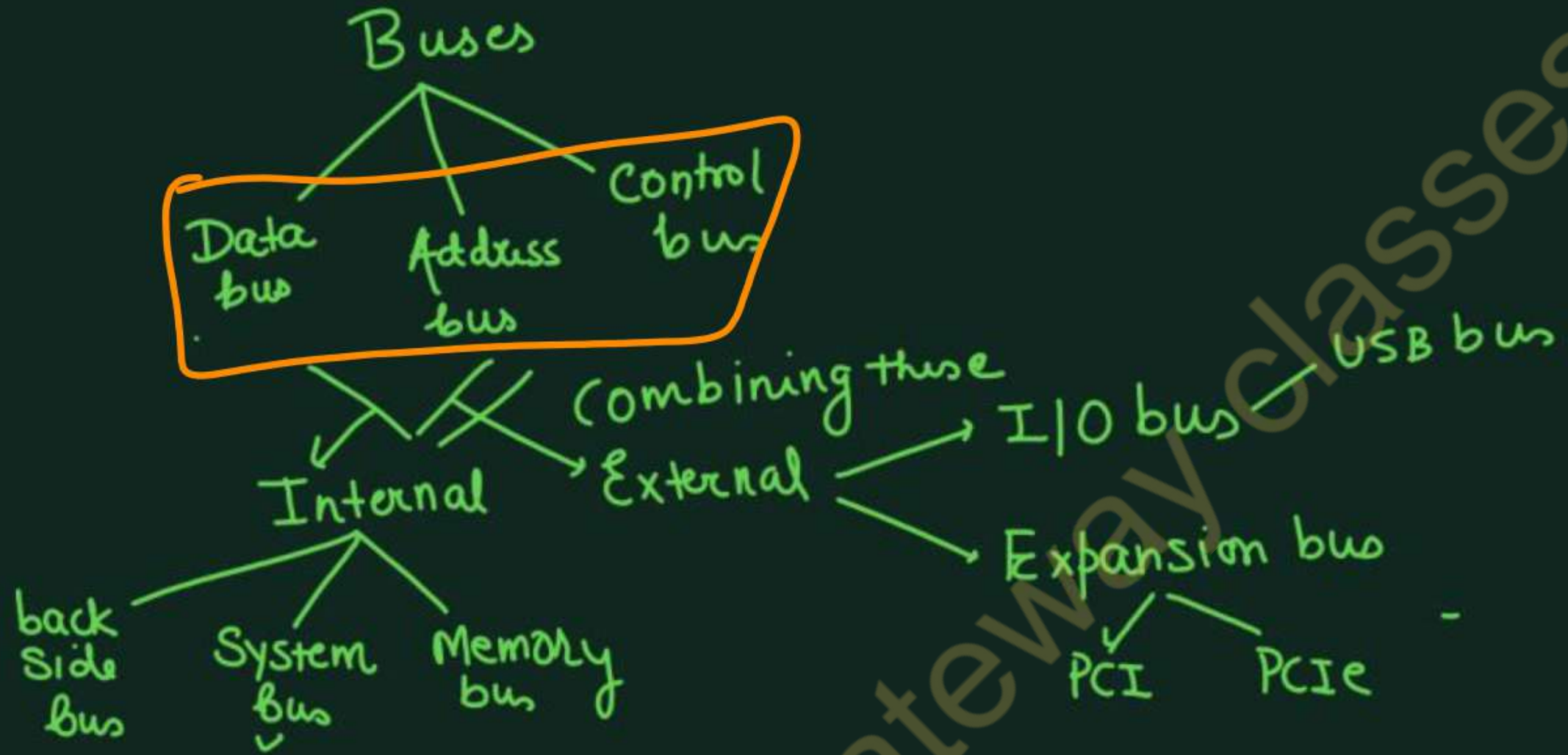
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Data Bus Width

- The width of a data bus refers to the number of bits (electrical wires) that the bus can carry at a time.
- Each line carries 1 bit at a time. So, the number of lines in data bus determine how many bits can be transferred parallelly.
- A 32-bit bus has thirty two (32) wires and thus can transmit 32 bits of data at a time.

Address Bus Width

- The width of address bus determines the amount of physical memory addressable by the processor.
- In other words, it determines the size of the memory that the computer can use.
- The wider is the address bus, the more memory a computer will be able to use, ^{but}
- An address bus that consists of 16 wires can convey 2^{16} (= 64K) different addresses.



1 internal bus

- An internal bus refers to the pathways within a computer that connect the internal components, such as the CPU, memory (RAM), and internal storage devices.

2.External bus

- It facilitates communication between the computer and external hardware, such as

keyboards, ^(mouse) mice, printers, and storage devices.

2.1. Input/Output (I/O) Bus:

- Connects external devices to the computer and manages data transfer between the computer and peripherals, such as USB and serial ports.
- Universal Serial Bus (USB), FireWire (IEEE 1394) (Keyboard, Mouse, etc)

2.2.Expansion Bus:

- Allows for the addition of extra hardware components to the computer, such as graphics cards and network cards, through slots on the motherboard. Examples include PCI

TYPES OF BUSES

Type of Bus	Function	Direction	Components Connected	Characteristics
<u>Data Bus</u>	Transfers actual data between components (CPU, memory, I/O devices).	<u>Bidirectional</u>	CPU, RAM, I/O devices.	Width varies (e.g., 8, 16, 32, 64 bits). Wider bus allows more data transfer at once.
<u>Address Bus</u>	Transfers memory addresses from the CPU to other components to specify where data is stored or retrieved.	<u>Unidirectional</u> (from CPU)	CPU, RAM, I/O devices.	Width determines maximum memory addressing capability (e.g., 16-bit, 32-bit).

TYPES OF BUSES

Type of Bus	Function	Direction	Components Connected	Characteristics
Control Bus	Transfers <u>control signals</u> from the <u>CPU</u> to <u>manage and coordinate operations</u> (e.g., <u>read/write commands</u> , <u>interrupt signals</u>).	<u>Bidirectional.</u>	<u>CPU, RAM, I/O devices, system clock.</u>	<u>Carries signals like read/write, interrupt requests, and clock pulses.</u>
PCIe Bus PCI (Older version)	Connects <u>high-speed components like graphics cards, sound cards, and network cards</u> to the <u>motherboard.</u>	<u>Bidirectional</u> (from CPU).	<u>CPU, GPU, sound cards, network cards.</u>	PCIe (<u>Peripheral Component Interconnect Express</u>) offers multiple lanes for data transfer, allowing faster communication.

TYPES OF BUSES

Type of Bus	Function	Direction	Components Connected	Characteristics
<u>Memory Bus</u>	Specifically connects the <u>CPU</u> and <u>memory</u> , including RAM and cache.	Bidirectional	CPU, RAM, cache.	High-speed bus designed to handle large volumes of data between CPU and memory.
<u>Back-Side Bus (BSB)</u>	Connects the <u>CPU</u> to the <u>Level 2 (L2) cache</u> (and sometimes <u>L3 cache</u>).	Bidirectional	CPU, L2/L3 cache.	Operates at a <u>higher speed</u> to <u>reduce latency</u> in <u>accessing cache memory</u> .

TYPES OF BUSES

Type of Bus	Function	Direction	Components Connected	Characteristics
USB Bus ✓	Connects external devices like <u>keyboards</u> , <u>mice</u> , <u>storage devices</u> , and <u>printers</u> to the computer.	Bidirectional ✓	External devices (keyboards, mice, storage). ✓	Universal Serial Bus (USB) standardizes connection for external devices, providing both data transfer and power supply. ✓

Why data bus is bi-directional?

- Because it allows the CPU to both read data from and write data to memory and peripherals.
- Data flows from the CPU to devices or from devices to the CPU, depending on the operation

Why control bus is bi-directional?

- The control bus is bidirectional because it carries control signals from the CPU to devices (e.g., read/write commands) and receives signals back from devices (e.g., acknowledgments or interrupts).

Why address bus is uni-directional?

- Because it only sends memory or I/O addresses from the CPU specifying where data should be read from or written to, without requiring any return signal

REGISTER

- A register is a small and temporary storage unit inside a computer's (CPU).
- It plays a vital role in holding the data required by the CPU for immediate processing.
- It usually holds a limited amount of data ranging from 8 to 64 bits, depending on the processor architecture.

Register Transfer :

The information transformed from one register to another register.

What are advantages of using many register?(AKTU 2018-19)

- When more registers are available, the CPU can store intermediate results and frequently accessed data directly in the registers instead of repeatedly reading and writing from memory.
- Having many registers allows the CPU to handle complex computations and multiple operations simultaneously.

DIFFERENT TYPES OF REGISTER

Memory Data Registers (MDR):

- It temporarily holds data that is being transferred to or from the memory. It acts as a buffer between the memory and the processor.

Accumulator:

- This is the most frequently used register used to store the intermediate results of arithmetic and logical operations.

Memory Address Registers (MAR):

- It holds the address of the location to be accessed from memory.
- MAR and MDR together facilitate the communication of the CPU and the main memory.

General Purpose Registers:

- These are numbered as $R_0, R_1, R_2, \dots, R_{n-1}$, and used to store temporary data during any ongoing operation.
- Its content can be accessed by assembly programming.

DIFFERENT TYPES OF REGISTER

Program Counter (PC)

- It contains the memory address of the next instruction to be fetched.

Instruction Register (IR):

- The IR holds the instruction which is just about to be executed.



Stack Pointer (SP)

3 061L

- The stack pointer points to the top of the stack, which is a part of the memory used to store function calls and other operations.

Flag register(status register or condition code)

- It is a special type of register in a computer's central processing unit (CPU) used to indicate the status of the CPU or the outcome of various operations such as Zero Flag, Carry flag, Sign Flag, Overflow Flag, Parity Flag, Auxiliary Carry Flag, and Interrupt Enable Flag.

BUS ARBITRATION (AKTU 2019-20/21-22/23-24)

- In a computer system, **multiple devices**, such as the CPU, memory, and I/O controllers, are connected to a common communication pathway, known as a bus.
- In order to transfer data between these devices, they need to have access to the bus.
- **Problem arise when multiple device try to access bus simultaneously**
- It can lead to data corruption and system instability

- Bus arbitration is the process of resolving conflicts that arise when multiple devices attempt to access the bus at the same time.

NOTE:

- The ^{device}controller that has access to a bus at an instance is known as a Bus master.
- The bus arbiter decide who will be the current bus master

Why bus arbitration is needed?

Preventing Data Collisions	When multiple devices attempt to use the bus simultaneously, data collisions can occur. Bus arbitration manages access to prevent these collisions.
Fair Access	Ensures that all devices get a <u>fair opportunity</u> to access the bus, preventing any single device from <u>monopolizing</u> the bus.
Efficient Resource Utilization	<u>Maximizes the utilization of the bus by coordinating access</u> , thereby improving <u>overall system performance</u> .
Priority Management	Allows for prioritization of bus access, giving <u>critical devices</u> or <u>high-priority tasks</u> <u>preferential access to the bus</u> .

Why bus arbitration is needed?

Conflict Resolution	Resolves conflicts that arise when multiple devices request access to the bus at the same time
Scalability	Supports <u>system expansion</u> by managing access as more devices are added, ensuring the bus can <u>handle increased traffic..</u>
Power Efficiency	Optimizes <u>power usage</u> by <u>controlling when and how devices access the bus</u> , potentially reducing power consumption.

How does the computer organisation

affect the performance of the computer?

- Computer organization affects performance through its architecture, including the CPU, memory hierarchy, (AKTU)
- Efficient design can reduce bottlenecks, improve processing speed, and enhance overall system throughput.
- For example, faster CPUs, larger caches, and optimized bus architectures can significantly boost performance.

Discuss the storage component of CPU (AKTU 2018-19)

- The primary storage component of a CPU is the register.
- Registers are small, fast storage locations within the CPU used to hold data and instructions that are currently being processed.

Q.1 ✓	What is bus arbitration? Why bus arbitration is needed ?	AKTU 2023-24 AKTU 2019-20/18-19 AKTU 2021-22
Q.2 ✓	What is system bus? Explain the different types of buses (D, A, IC) Or What are the different types of buses in computer architecture?	AKTU 2016-17/17-18 AKTU 2021-22/2023-24
Q.3 ✓	What are advantages of using many register?	AKTU 2018-19
Q.4 ✓	How does the computer organisation affect the performance of the computer?	(AKTU 2022-23)
Q.5 ✓	What is system bus? Explain different types of bus.	AKTU (2011-12)
Q.6 ✓	Explain the role of <u>program counter</u> and <u>instruction register</u> .	AKTU (2022-23)/(2015-16)
Q.7 ✓	Explain arithmetic and logic unit.	AKTU(2021-22)

Computer organization and architecture

UNIT 1 INTRODUCTION

Today's Target

- **Bus arbitration types**
- **Decoder and multiplexer**
- **Memory transfer and bus transfer**
- **AKTU PYQs**

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There are two approaches to bus arbitration:

Centralized bus arbitration –

One controller, called the bus arbiter, decides which device gets to use the bus when multiple devices request it.

Distributed bus arbitration –

All devices participating in the selection of the next bus master.

Methods of Centralized BUS Arbitration:

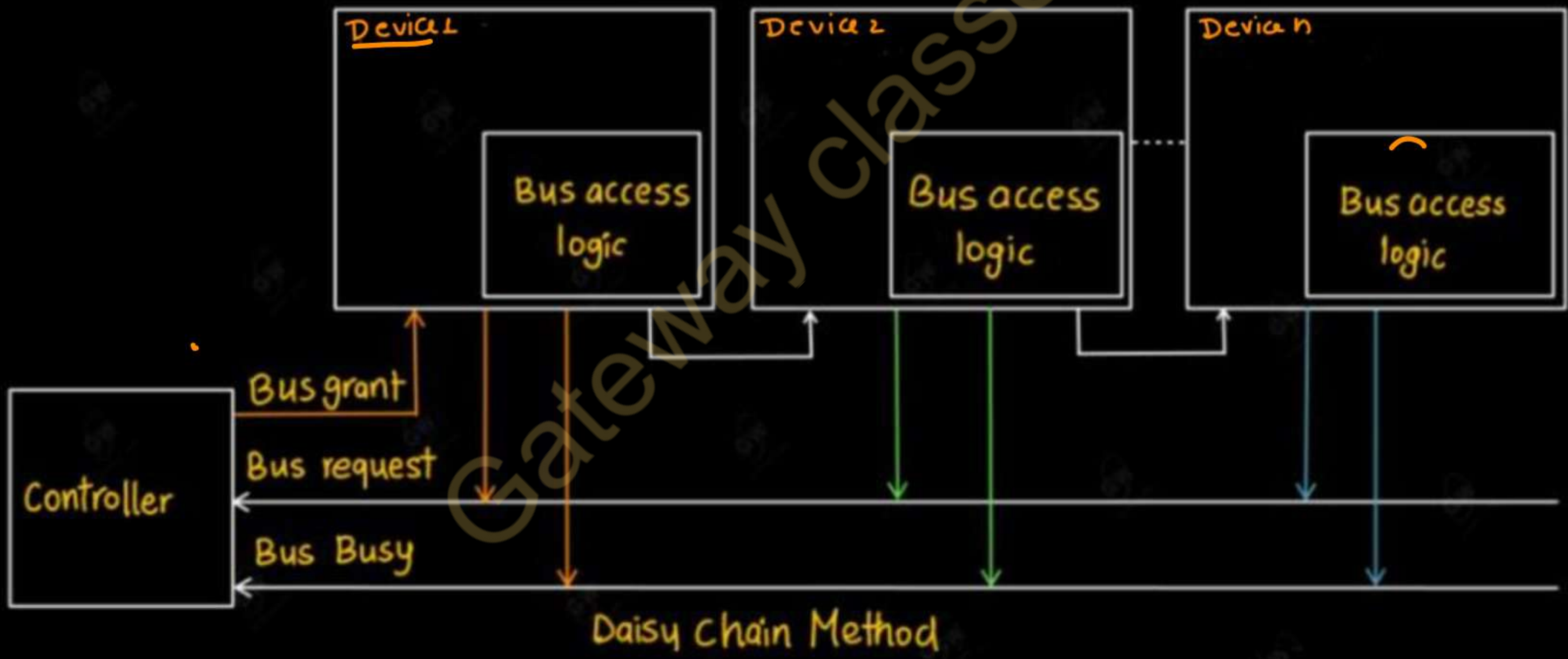
- Daisy Chaining method.
- Polling or Rotating Priority method.

➤ Fixed priority or Independent Request method

1. Daisy Chaining method

- The same line is used for bus requests by all devices.
- The Bus Controller sends out the Bus Grant signal if the Bus Busy line is not active.
- Beginning with the nearest device, the bus grant signal is transmitted serially through every device.
- The device, who needs the System Bus, get the control of the System Bus and then Bus Busy line is activated and become bus master.

DAISY CHAINING



Advantage:

- Simple design is an advantage.
- There are fewer control lines.
- It is also simple to add new device.

Disadvantage:

- The device who is physically closest to the bus arbiter has the highest priority.
- Bus access is granted serially, which causes a propagation delay in the circuit.
- If one device stops working, the whole system might stop working too.

2. Polling or Rotating Priority method

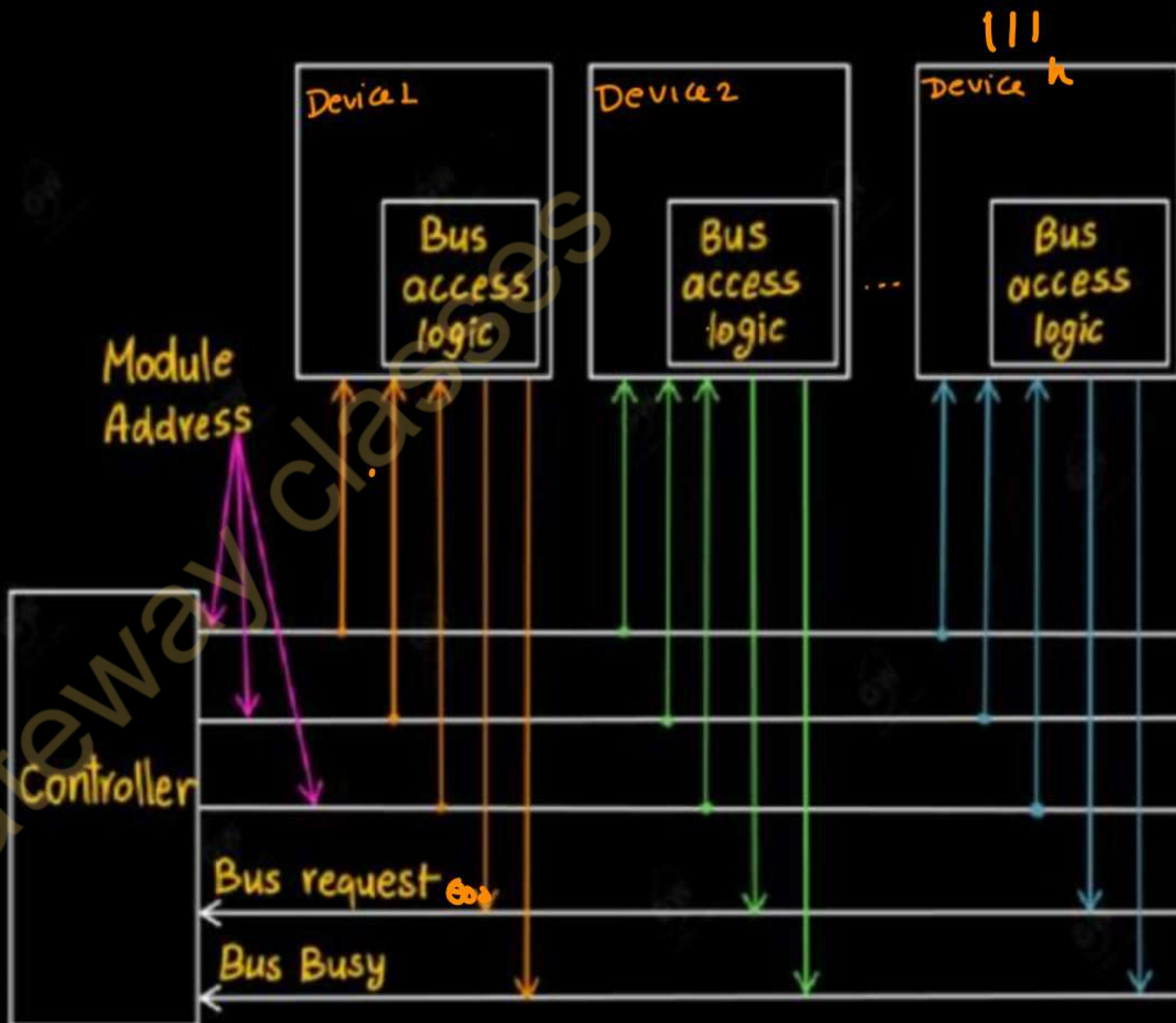
- The same line is used for Bus Request by all device in this instance. Here, the controller creates a master's binary address.
- For instance, 3 address lines are required ($2^3 = 8$) to connect 8 bus device. The controller "polls" the device in response to a bus request by transmitting a series of bus master addresses on the address lines.
- The selected device takes control of the bus and signals that the bus is in use. (Bus busy line is activated)

Advantages

- This method does not favor any particular device and processor.
- The method is also quite simple.
- If one device fails then the entire system will not stop working.

Disadvantages –

- Adding new device is difficult as increases the number of address lines of the circuit.



Fixed priority or Independent Request method

- Each device has a separate Bus Request and Bus Grant here.
- By knowing which device has asked, the controller can give that device the bus.
- If multiple devices request the bus at the same time, the bus is given to the one with higher priority, as long as the bus is not already in use. (Bus busy line is not activated)

- The controller is made up of circuits that determine the order of priority using encoder and decoder logic

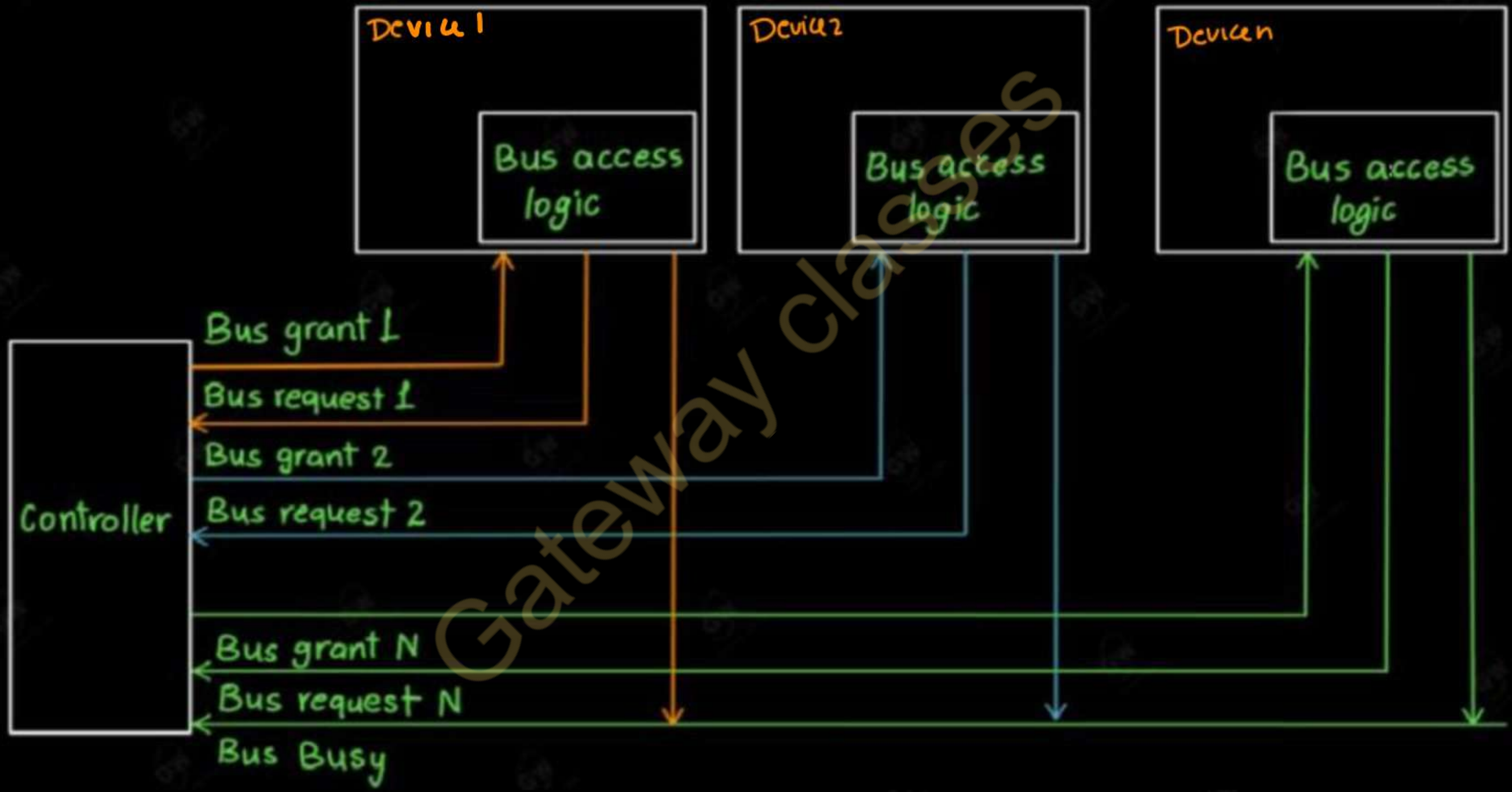
✓ Advantage:

- This technique produces a quick response.
- The speed of Bus Arbitration is unaffected by the quantity of connected devices.

➤ Disadvantage:

- More control lines are needed (2^n lines are needed for n devices).

Independent Requests



NOTE

Imp

PCIe Buses

→ Bidirectional bus.

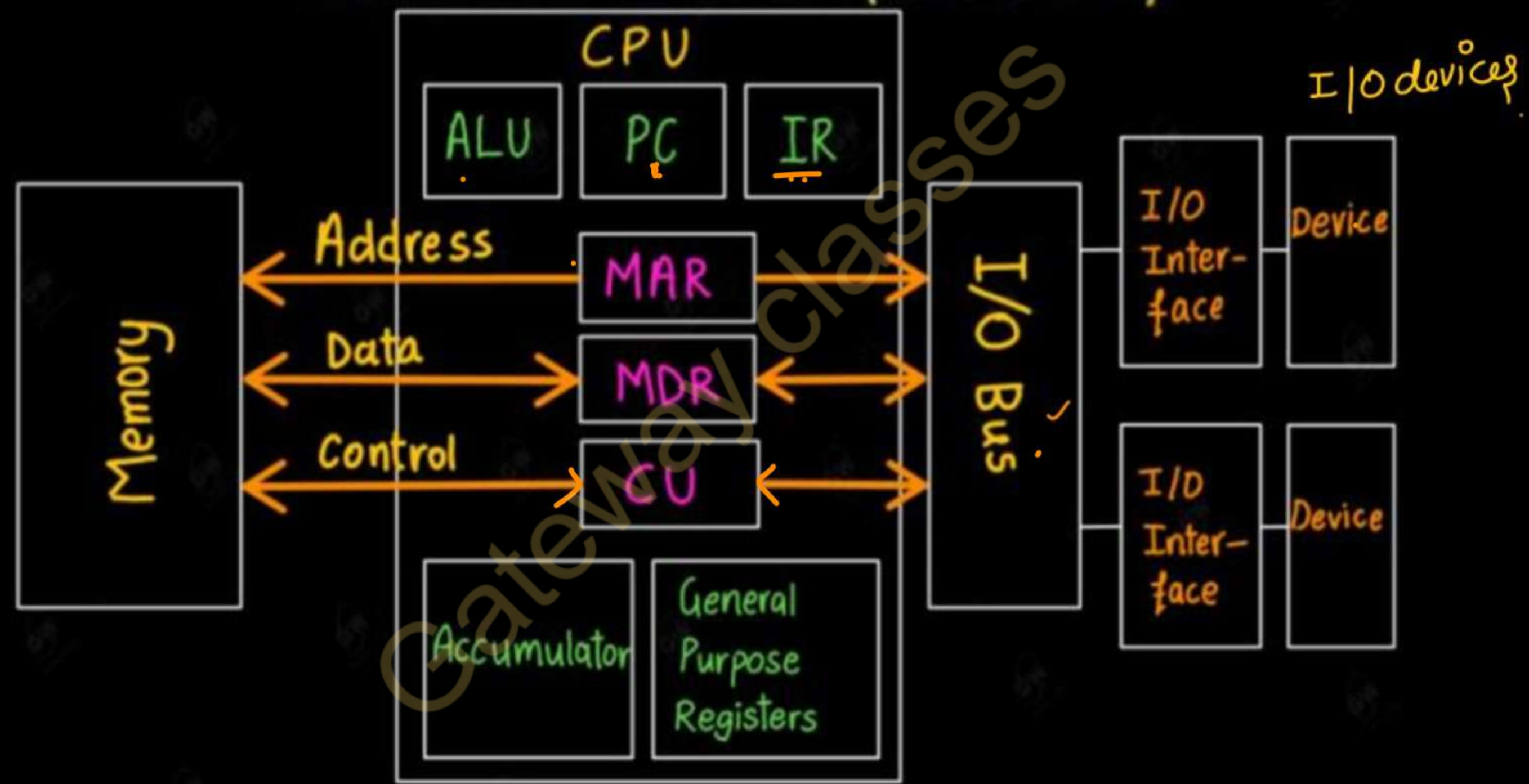
David A. Patterson

John L. Hennessy

Computer Organisation

System bus
bidirectional

BLOCK DAIGRAM of CPU (AKTU 2016-17)



1. Memory Address Register (MAR)

➤ It holds the memory address of the data that needs to be accessed, either for reading from or writing to memory.

2. Memory Data Register (MDR)

➤ This is a CPU register that temporarily holds data being transferred to or from memory.

➤ When the CPU reads from or writes to memory, the data passes through the MDR.

3 Program Counter (PC)

➤ It is a register in the CPU that holds the address of the next instruction to be executed in a program.

4. The Instruction Register (IR)

➤ It is a CPU register that holds the current instruction being executed. After the CPU fetches an instruction from memory, it is stored in the IR.

5. Input-Output (I/O) bus

- It is a communication pathway that connects the CPU and memory to peripheral devices, such as keyboards, printers, and storage drives. (Pendrives)
- It facilitates data transfer between the central system and these external devices.

6. Accumulator

- The Accumulator is a special register within the CPU used to store intermediate

results of arithmetic and logic operations.

7 I/O interface

- An I/O (Input/Output) interface is a system that facilitates communication between the CPU and peripheral devices.
- It acts as a bridge, converting data and control signals from the CPU into a form that peripheral devices can understand, and vice versa

8 A General-Purpose Register

➤ It is a CPU register used for storing temporary data and performing arithmetic or logical operations.

➤ It can hold intermediate results and data for various tasks, providing versatility in execution.

Its use helps optimize processing efficiency by minimizing memory access.

Address bus, data bus, Control bus, control unit,

ALU(Already explained)

MULTIPLEXER(MUX) 2 Marks

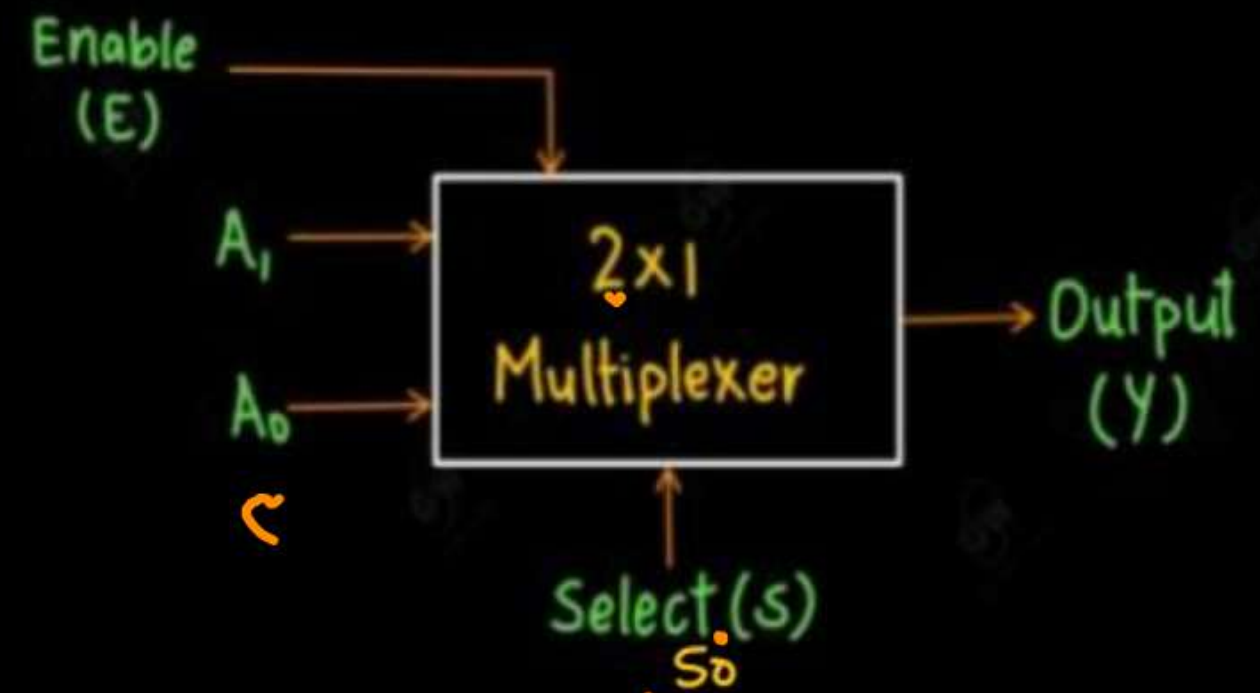
❑ A multiplexer is a combinational circuit that has 2^n input lines and a single output line.

❑ It is a multi-input and single-output combinational circuit.

2x1 mux

$$2 \text{ I/P} \rightarrow \log_2 2 = 1$$

$n = \text{Select line}$



INPUTS		OUTPUT
S0		Y
0		A0
1		A1

E	I/P	O/P
E	S0	Y
X	—	—
L	0	A0
L	1	A1

4X1 MUX

INPUTS		OUTPUT
S1	S0	Y
0	0	A0
0	1	A1
1	0	A2
1	1	A3



DECODER

The combinational circuit that change the binary information into 2^N output lines is known as Decoders.



ENABLE	INPUTS		OUTPUTS			
E	A_0	A_1	Y_3	Y_2	Y_1	Y_0
0	X	X	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

Memory Transfer/

HOW MEMORY READ AND WRITE

OPERATION PERFORMED (AKTU 2022-23)

- The transfer of information from a memory unit to the user end is called a Read operation.
- The transfer of new information to be stored in the memory is called a Write operation.
- A memory word is designated by the letter **M**.
- We must specify the address of memory word while writing the memory transfer operations.
- ☐ The address register is designated by **AR** and the data register by **DR**.

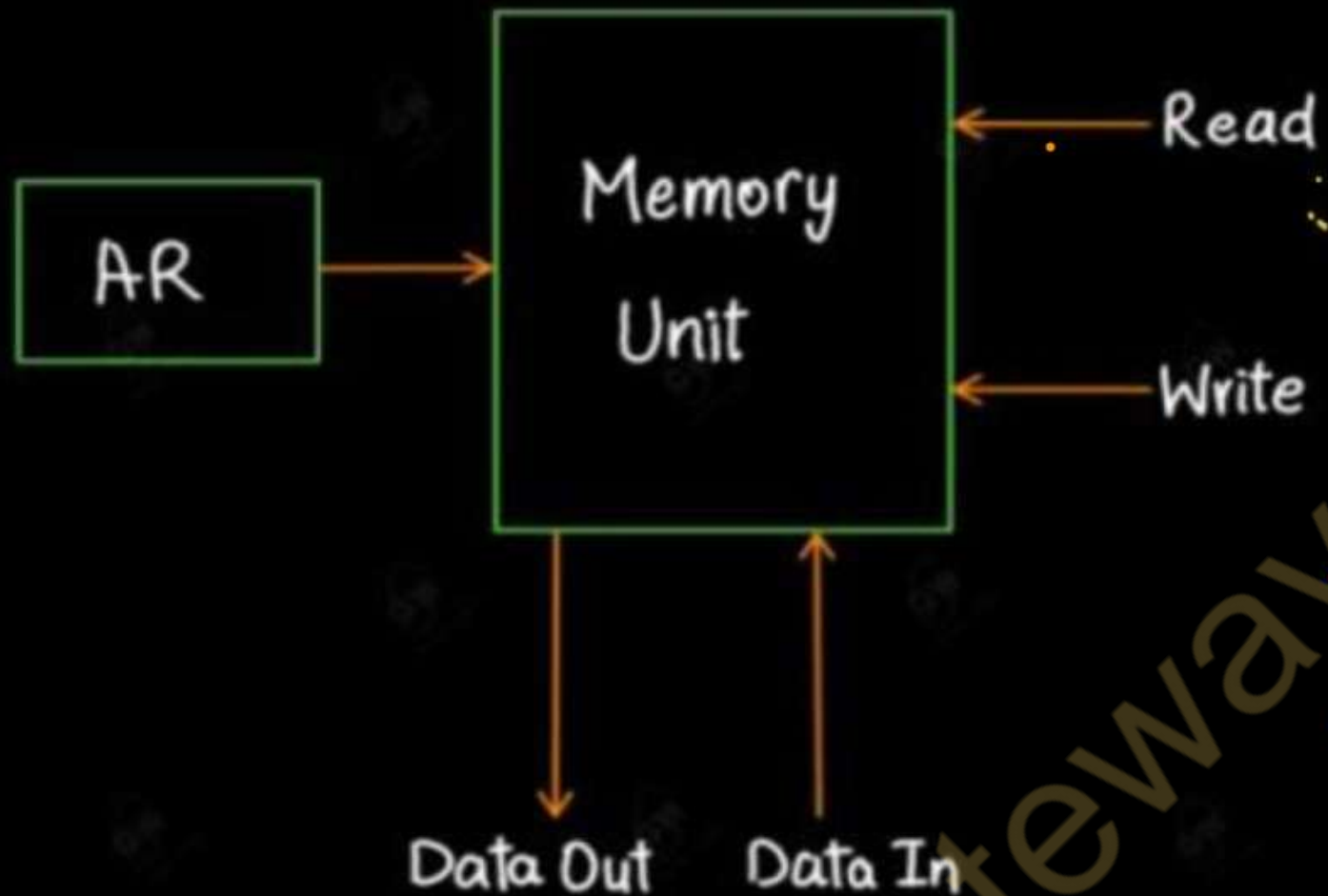
The address register and data register mainly used for memory transfer.

Read: $DR \leftarrow M[AR]$

- The Read statement causes a transfer of information into the data register (DR) from the memory word (M) selected by the address register (AR).

Write: $M[AR] \leftarrow R1$

- The Write statement causes a transfer of information from register R1 into the memory word (M) selected by address register (AR).



Bus Transfer:

The act of sending data from one component to another through the bus. This can involve reading from or writing to memory, or exchanging data between devices.

There are two methods in bus transfer:

- Bus transfer using multiplexer.
- Bus transfer through three state buffer.

AKTU Questions

Q1	What is <u>bus</u>? Define bus and memory transfer.	AKTU 2022-23 AKTU 2018-19
Q2	What is <u>bus arbitration</u>? List different types of bus <u>arbitration</u> methods <i>Q Daisy Chaining & polling Advantage & Disadvantage</i> Or What is meant by the term <u>bus arbitration</u>? Why is it <u>needed</u>? How can bus arbitration be implemented in daisy changing scheme?	AKTU 2021-22 AKTU 2020-21 AKTU 2023-24
Q3	How memory read and write operations are performed in computer system?	AKTU 2022-23

Computer organization and architecture

UNIT 1

INTRODUCTION

Today's Target

- Bus transfer through multiplexer and three state buffer
- Addressing modes
- AKTU PYQs

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PDF notes are available in the App (Link in Description)

Bus transfer through multiplexer

Or

Common Bus System using multiplexers

- A digital system composed of many registers, and paths must be provided to transfer information from one register to another.

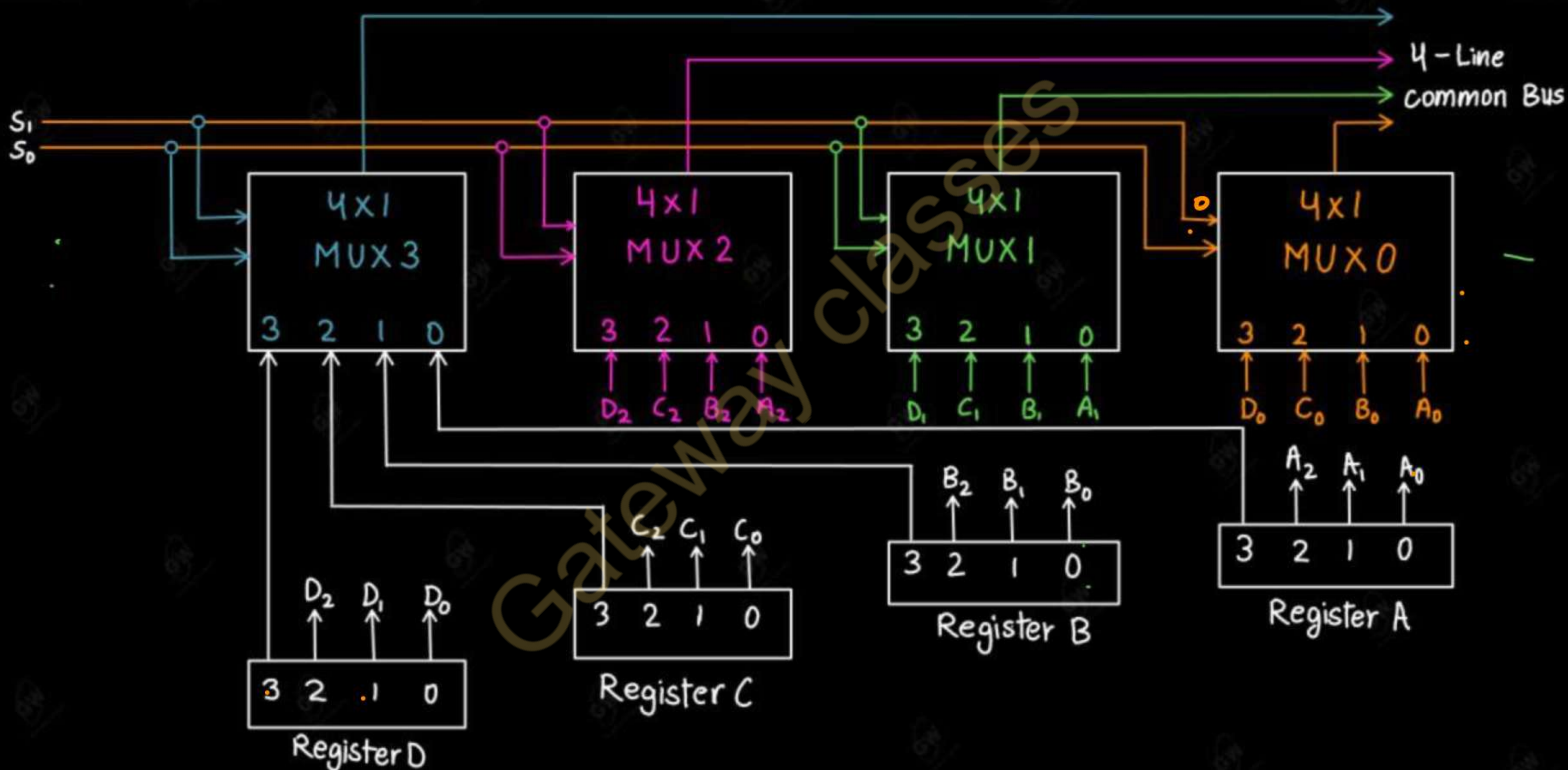
- The number of wires connecting all of the registers will be excessive if separate lines are used between each register and all other registers in the system.
- A bus structure, on the other hand, is more efficient for transferring information between registers in a multi-register configuration system.
- The two selection lines S1 and S2 are connected to the selection inputs of all four multiplexers. The selection lines choose the four bits of one register and transfer them into the four-line common bus.

When both of the select lines are at low logic, i.e. $S_1S_0 = 00$, the 0 data inputs of all four multiplexers are selected and applied to the outputs that forms the bus.

- This, in turn, causes the bus lines to receive the content of register A since the outputs of this register are connected to the 0 data inputs of the multiplexers.

S1	S0	REGISTER SELECTED
0	0	A
0	1	B
1	0	C
1	1	D

Implement the bus of 4 lines with MUX



NOTE:

1. Number of multiplexers needed =
number of bits in each register

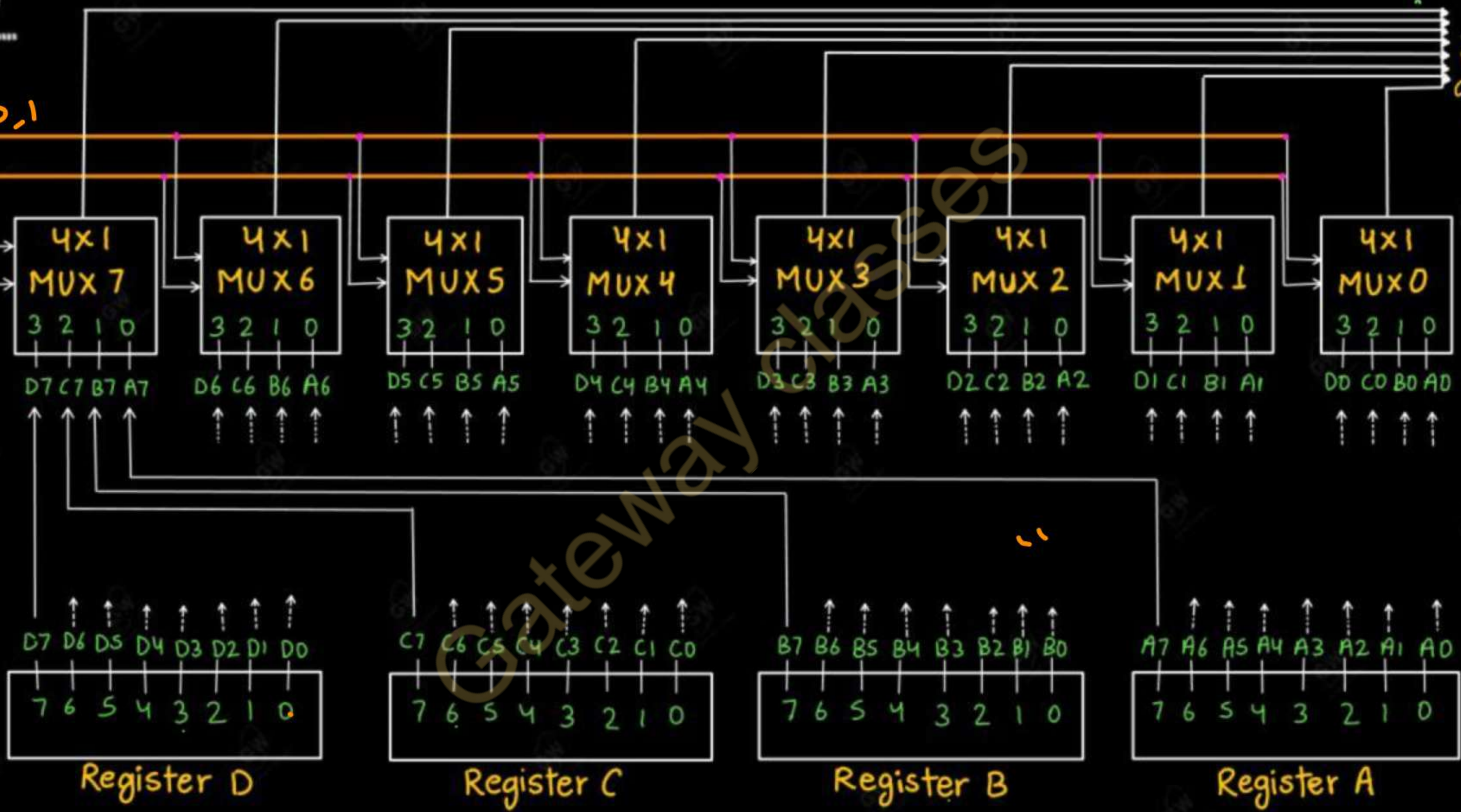
2. Number of inputs in each
multiplexer needed = Total number of
registers

Implement a bus of 8 lines with MUX

S1	S0	REGISTER SELECTED
0	0	A
0	1	B
1	0	C
1	1	D

SIO, I
S01

8-Line
Common
Bus



BUS TRANSFER THROUGH THREE STATE BUFFER

(BUS TRANSFER THROUGH THREE STATE BUFFER AND DECODER INSTEAD OF MULTIPLEXER)(AKTU 2020-21/2021-22)

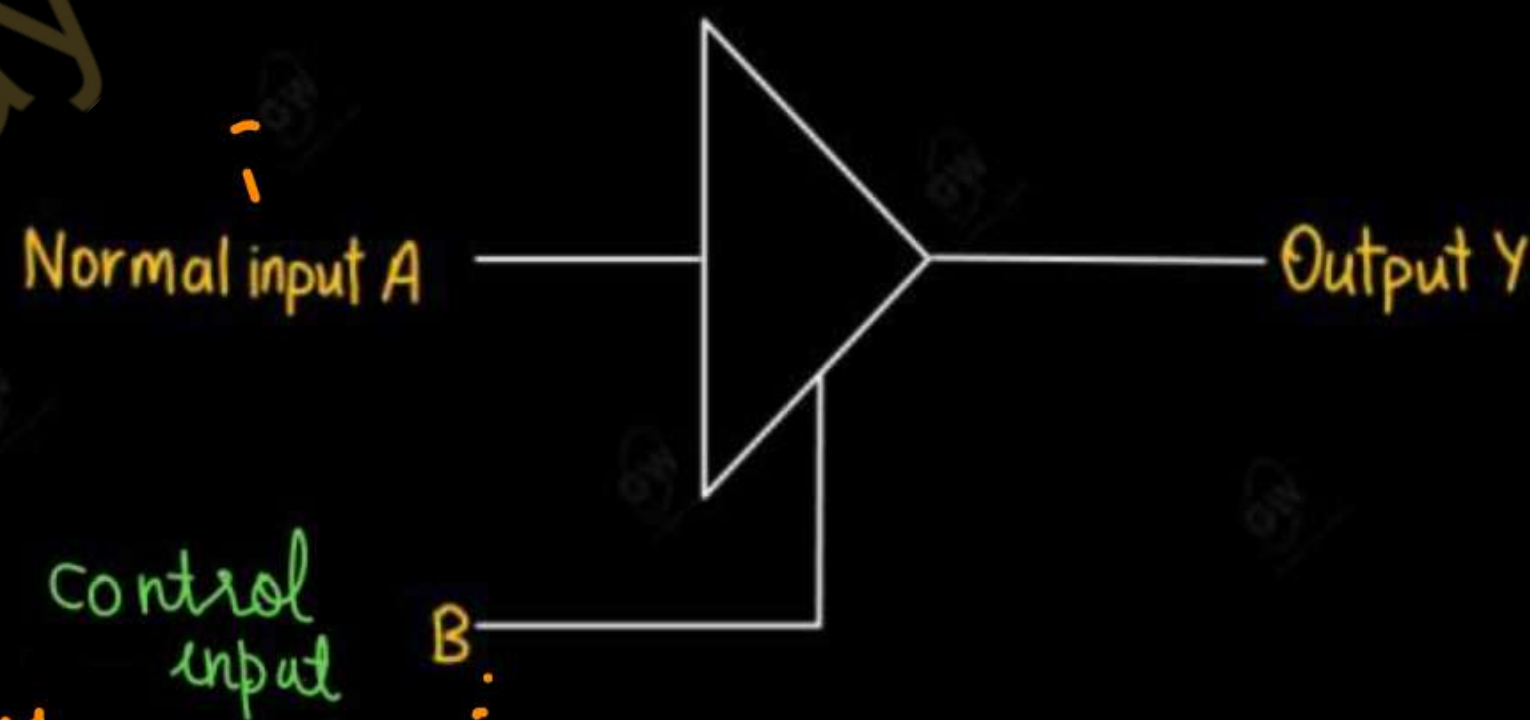
- The three state gates can be considered as a digital circuit that has three gates, two of which are signals equivalent to logic 1 and 0 as in a conventional gate. However, the third gate exhibits a high-impedance state.

- The most commonly used three state gates in case of the bus system is a buffer gate
- The graphical symbol of a three-state buffer gate can be represented as:

$Y=A$ if $B=1$

High impedance if $B=0$

$B=1$
 $A=0 \quad Y=0$
 $A=1 \quad Y=1$
 $B=0$



Three state gates have two inputs,

normal input and control input.

➤ Control input determines the output of the three state gates :

➤ If control input is equal to 1, then the output is the normal input in the gate.

➤ If control input is equal to 0, then the gate goes to a high impedance state (a state in which output is disconnected).

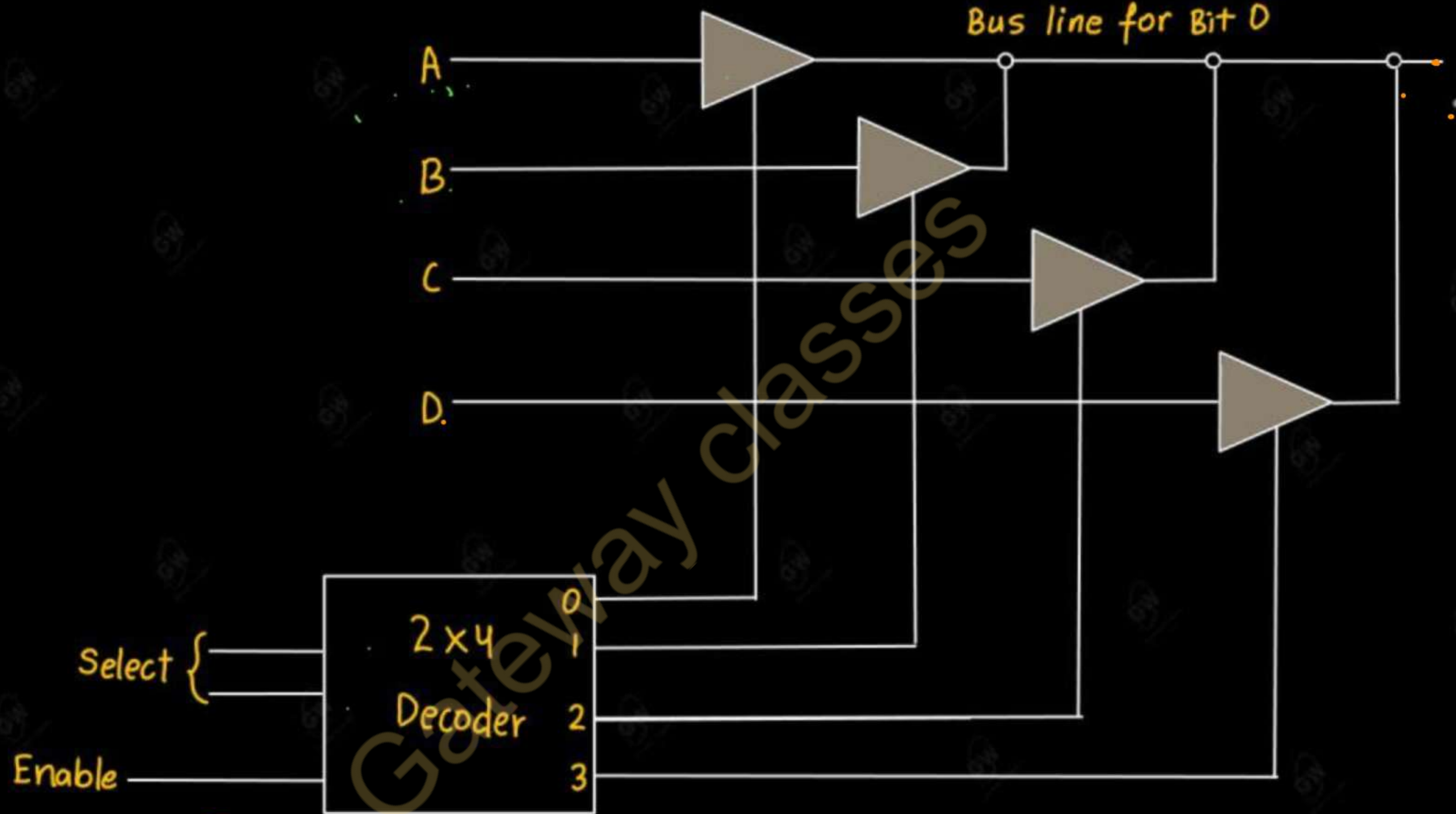
Working:

➤ Four three state gates are used for each bit whose outputs are connected together to form a single line of a common bus

➤ Each gate has control input which decides whose data will be transferred to the bus line.

The control input of only one gate is active at a time. The other gates are at high impedance state.

➤ To ensure that the control input of only single gate is active at a time, a decoder is used



Decoder has enable input :

- If the **enable input is 0**, then the **control inputs of all the gates are disabled** and the bus line goes to the high impedance state.
- If the **enable input is 1**, then the control input of any one of the gate is active depending on the **binary value of the selection variables**.

SOME IMPORTANT POINTS:

- The operation field(opcode) specifies the operation to be performed, like addition
- Address field which contains the location of the operand, i.e., register or memory location.
- Mode field which specifies how operand is to be founded.

MODE	OPCODE	ADDRESS FIELD
------	--------	------------------

(AKTU 2022-23/2019-20)

The different ways of specifying the location of an operand in an instruction

Types of Addressing Modes-

- Implied / Implicit Addressing Mode
- Stack Addressing Mode
- Immediate Addressing Mode
- Direct Addressing Mode
- Indirect Addressing Mode
- Register Direct Addressing Mode

- Register Indirect Addressing Mode
- Relative Addressing Mode
- Indexed Addressing Mode
- Base Register Addressing Mode
- Auto-Increment Addressing Mode
- Auto-Decrement Addressing Mode

The effective address refers to the location where the operand resides.

➤ Significance of addressing mode

Addressing modes specify how the operand of an instruction is accessed. They provide flexibility in accessing data, optimize memory usage, and improve instruction efficiency.

1. Implied Addressing Mode (implicit addressing mode.)

- The definition of the instruction itself specify the operands implicitly
- EXAMPLE: Complement Accumulator

2. Stack Addressing Mode

- The operand is contained at the top of the stack.
- EXAMPLE : ADD

3. Immediate Addressing Mode-

- The operand is specified in the instruction explicitly.

➤ EXAMPLE:

1. MOV R #20

2. ADD 10

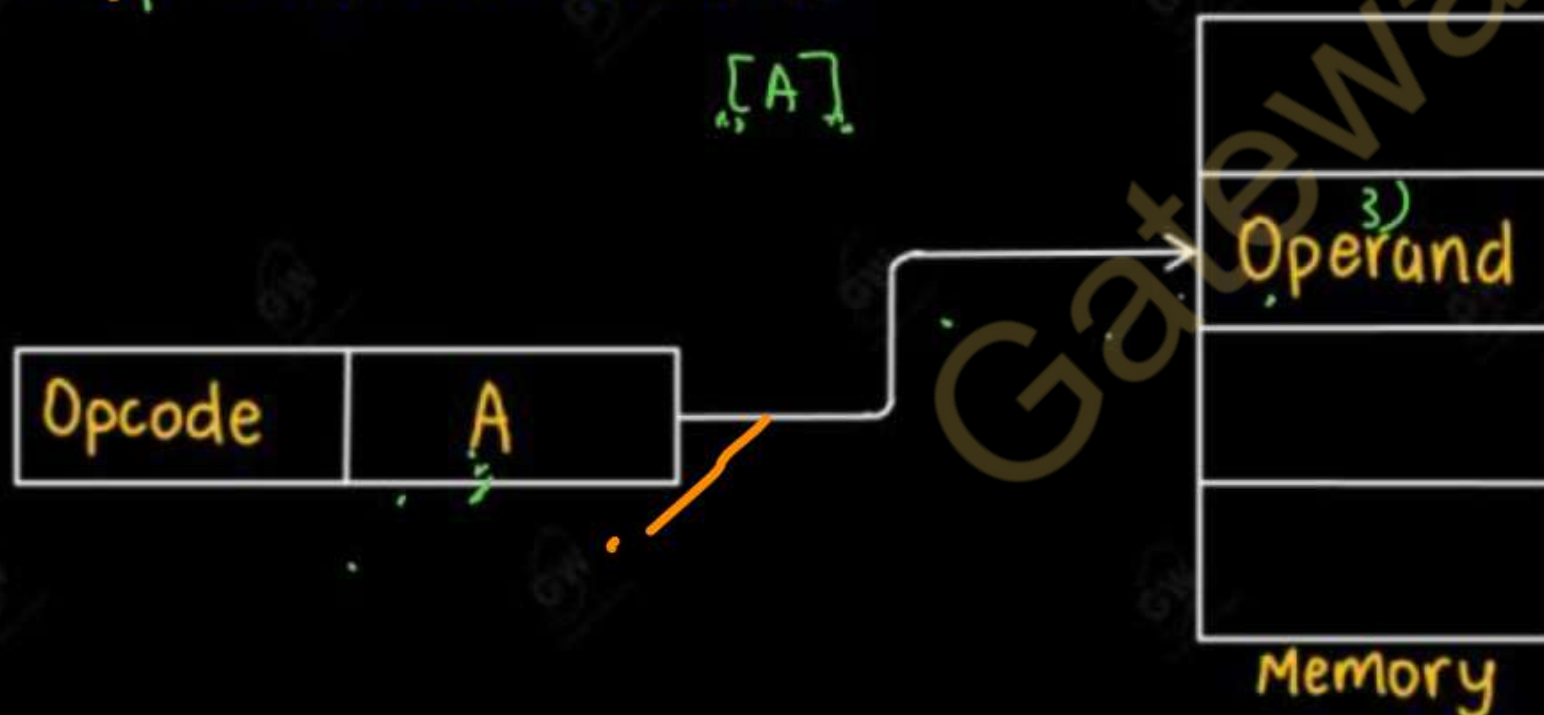
OPCODE

OPERAND

(Address part)

4. Direct Addressing Mode- (absolute addressing mode)

- The address field of the instruction contains the effective address of the operand.
- Only one reference to memory is required to fetch the operand.
- **ADD A ($AC \leftarrow AC + [A]$)**



5. Indirect Addressing Mode-

- The address field of the instruction specifies the address of memory location that contains the effective address of the operand.
- Two references to memory are required to fetch the operand.

AC $\leftarrow$ AC + [[A]]



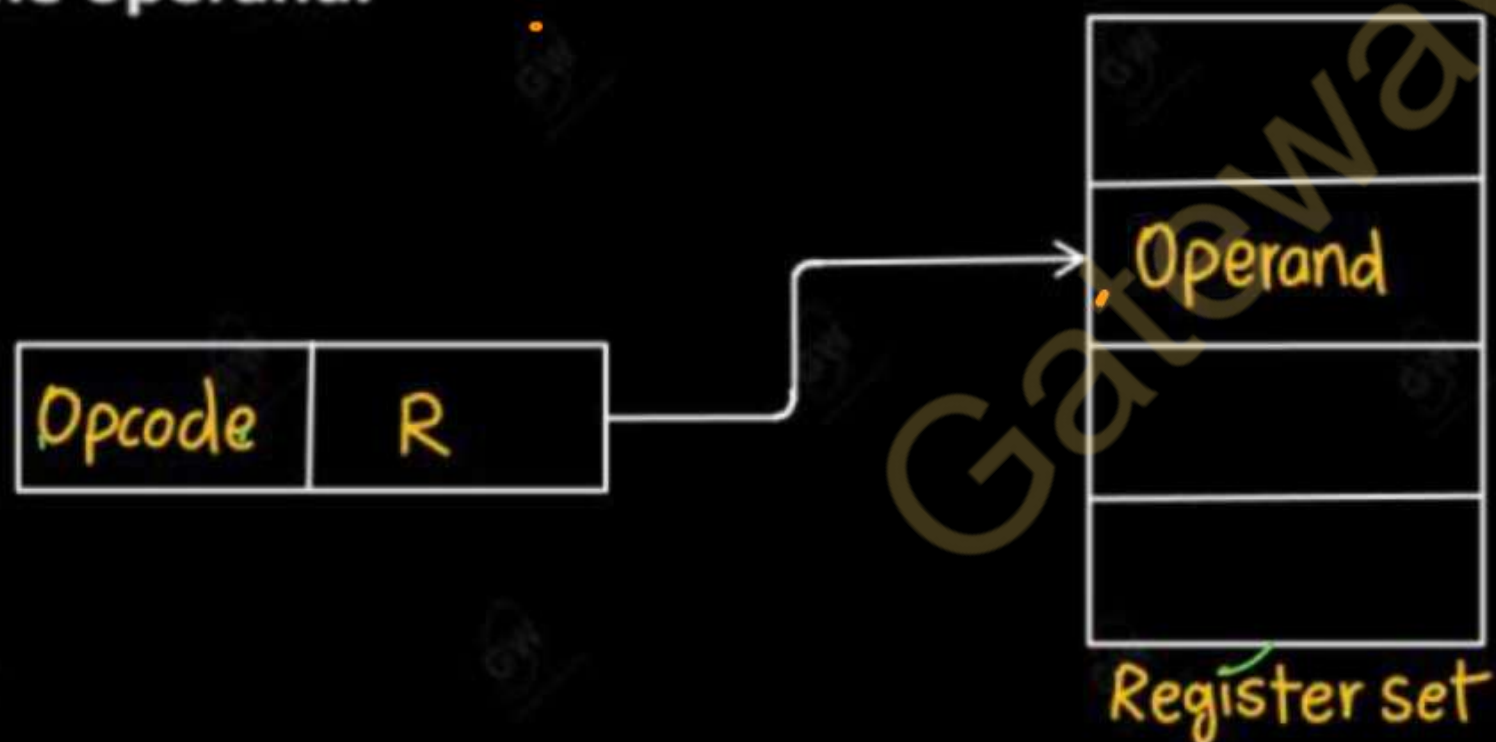
ADD 10
✓ ADD 10
ADD 10

$AC \leftarrow AC + 16$
 $AC \leftarrow AC + 3$ ①
 $ACK \leftarrow AC + 4$ ②

10	3EA
3	4

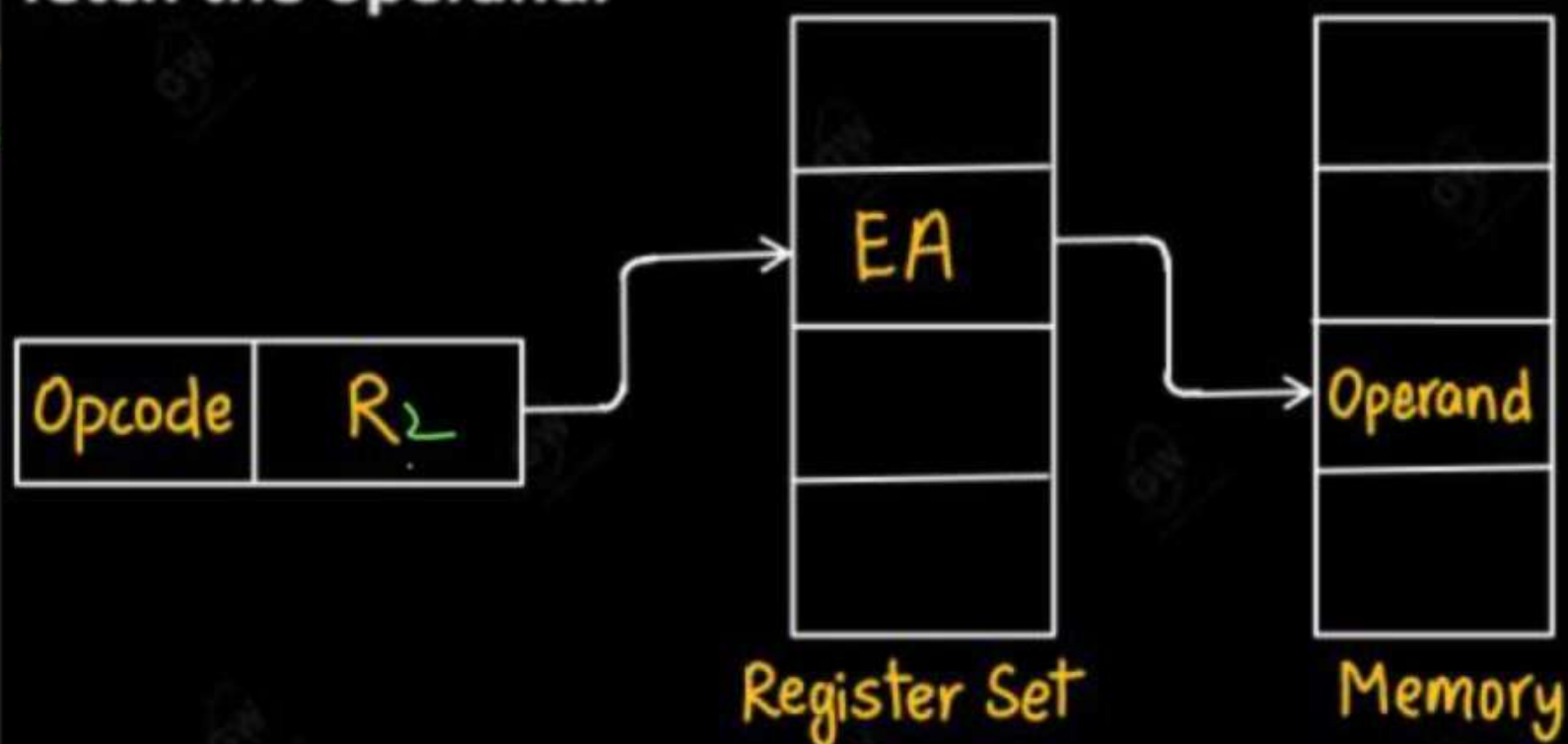
6. Register Direct Addressing Mode-

- The operand is contained in a register set.
- The address field of the instruction refers to a CPU register that contains the operand.
- No reference to memory is required to fetch the operand.



7. Register Indirect Addressing Mode

- The address field of the instruction refers to a CPU register that contains the effective address of the operand.
- Only one reference to memory is required to fetch the operand.



AKTU PYQs

Q.1	Illustrate <u>Bus system</u>? Implement a <u>bus of 4 lines with MUX</u>. <u>or</u> Draw a diagram of a bus system using MUX Which has four register of size <u>4 bits each</u>.	AKTU 2023-24 AKTU 2021-22
Q.2	What is significance of addressing modes? Explain various addressing modes. <u>or</u> Discuss <u>register indirect</u> and <u>memory indirect mode</u> of instruction with <u>example</u>.	AKTU 2023-24 AKTU 2019-20 AKTU 2018-19 AKTU2015-16
Q.3	Draw a diagram of a bus system In which its uses 3 states <u>buffer</u> and a decoder instead of multiplexer	AKTU 2020-21
Q.4	List three control signal.	AKTU2018-19

Computer organization and architecture

UNIT 1 INTRODUCTION

Today's Target

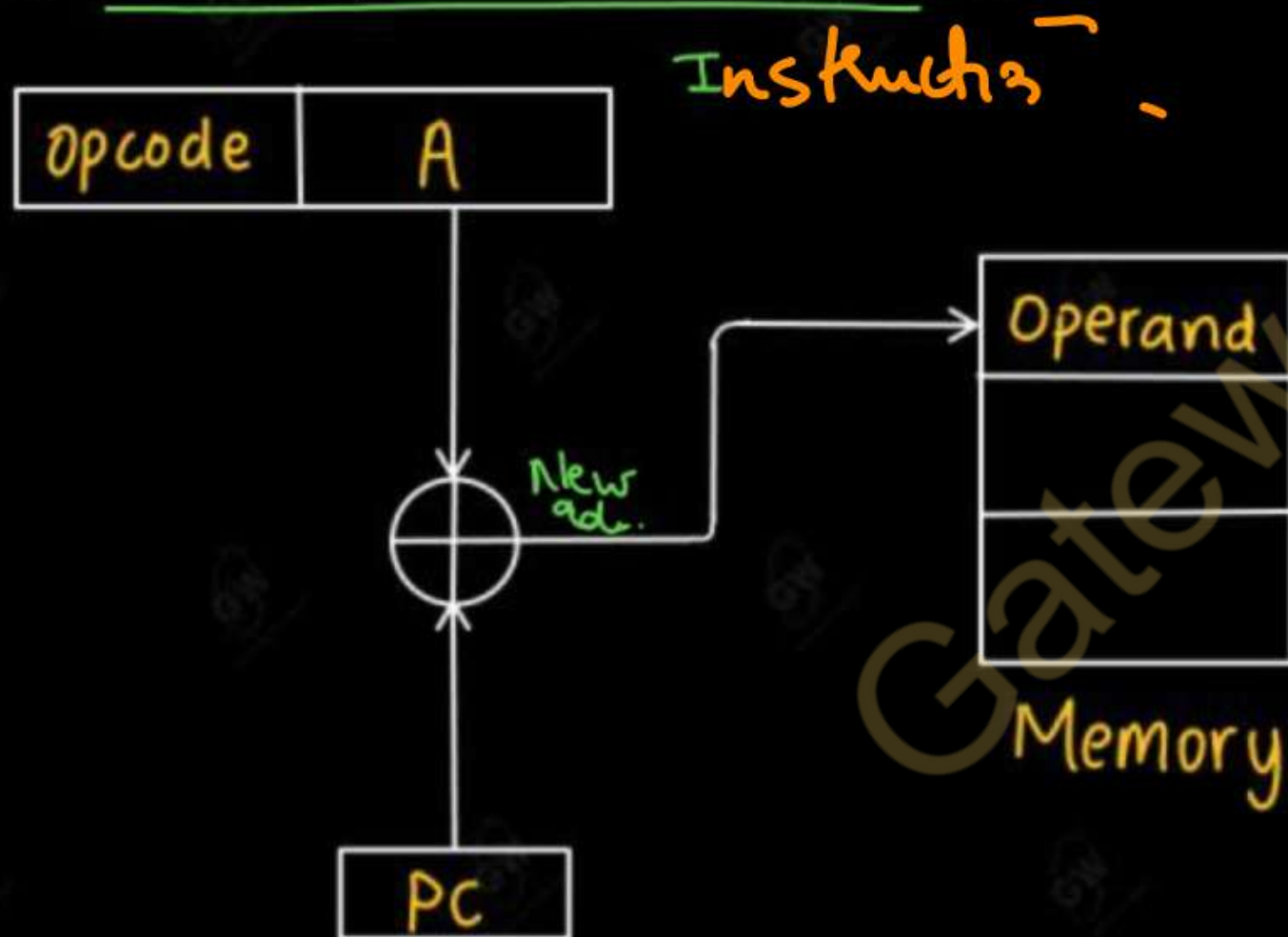
- Addressing modes
- Processor Organisation
- AKTU PYQs

By PRAGYA RAJVANSHI

B.Tech, M.Tech(C.S.E.)

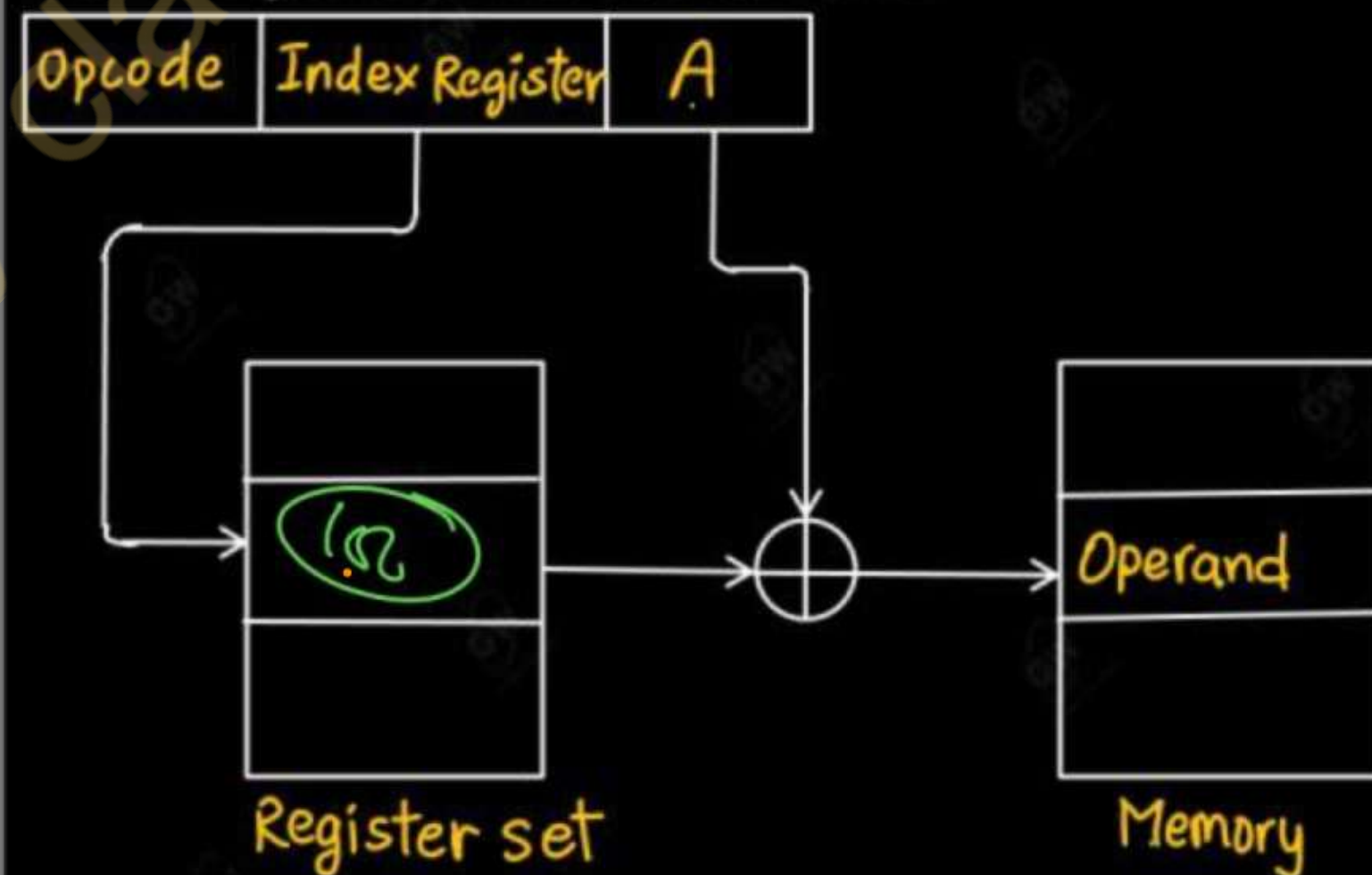
8. Relative Addressing Mode-

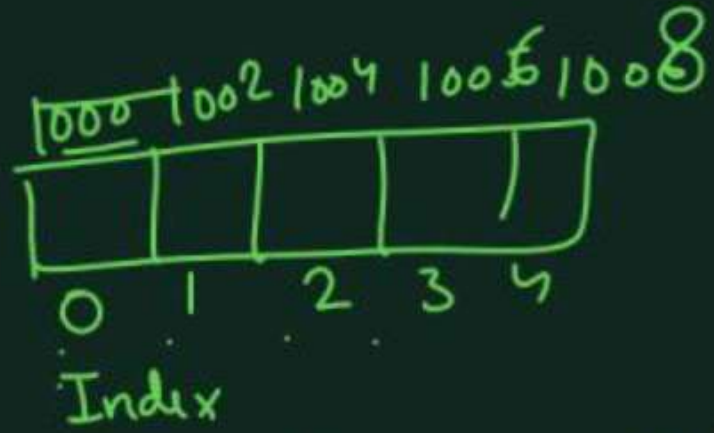
➤ Effective address of the operand is obtained by adding the content of program counter with the address part of the instruction.



9. Indexed Addressing Mode-

➤ Effective address of the operand is obtained by adding the content of index register with the address part of the instruction.



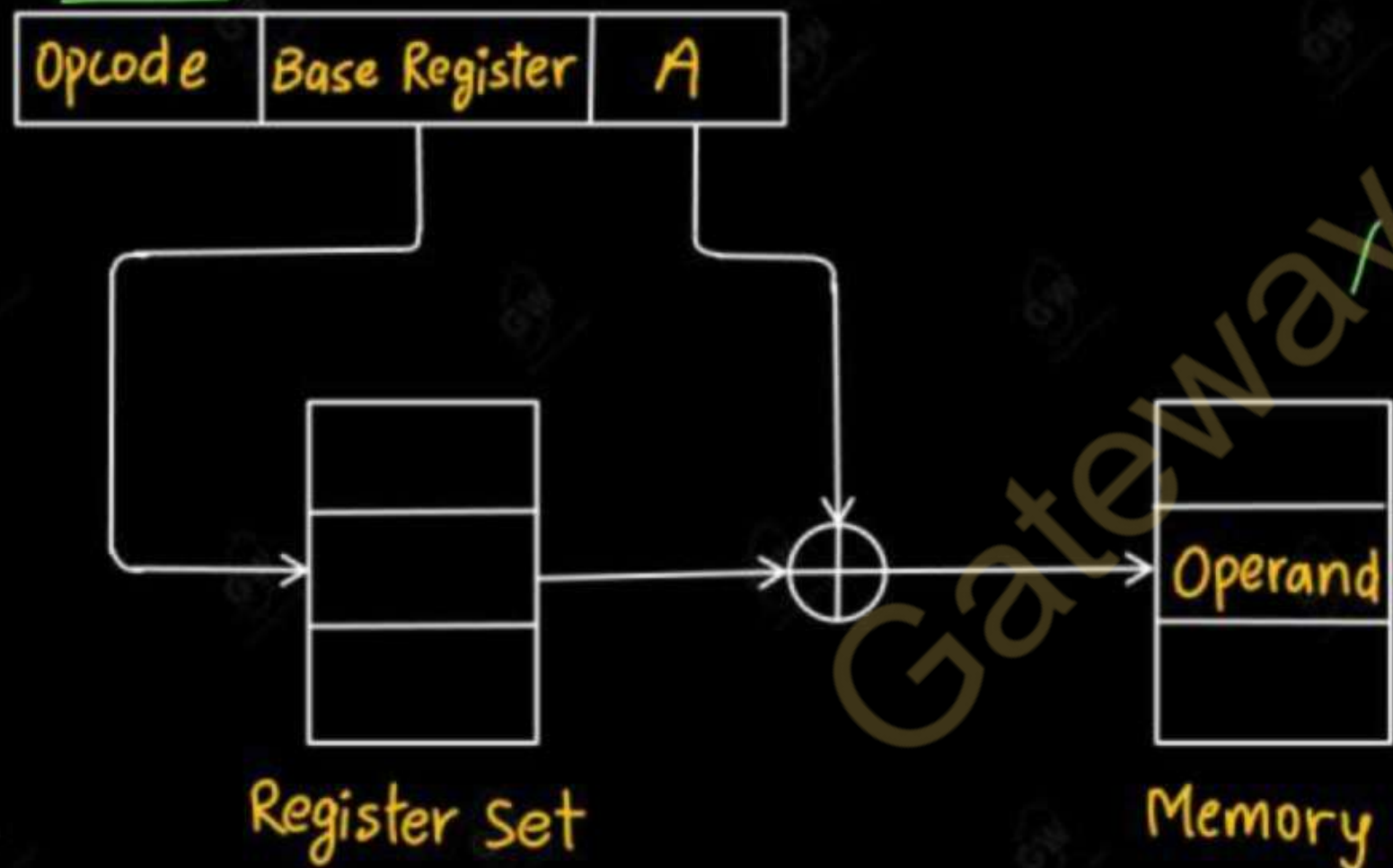


$$\boxed{1000} + 2 = 1002 + 2 = 1004$$
$$+ 2 = 1006$$

① step 81,

10. Base Register Addressing Mode

- Effective address of the operand is obtained by adding the content of base register with the address part of the instruction.



11. Auto-Increment Addressing Mode-

- This addressing mode is a special case of Register Indirect Addressing Mode

Effective Address of the Operand = Content of Register

- Only one reference to memory is required to fetch the operand.



11. Auto-Increment Addressing Mode-

- After accessing the operand, the content of the register is automatically incremented by step size 'd'.
- Step size 'd' depends on the size of operand accessed.

12 Auto-Decrement Addressing Mode-

- This addressing mode is again a special case of Register Indirect Addressing.

➤ Effective Address of the Operand = $\text{Content of Register} - \text{Step Size}$.

- First, the content of the register is decremented by step size 'd'.
- Step size 'd' depends on the size of operand accessed.
- After decrementing, the operand is read.
- Only one reference to memory is required to fetch the operand.



PROCESSOR ORGANISATION

(AKTU 2023-24/22-24/19-20)

- Processor organization refers to the **internal architecture** and **design of a CPU**, including how components like the **ALU** (Arithmetic Logic Unit), control unit, **registers**, and **buses** are arranged and interact to execute instructions.
- It defines the CPU's **data processing capabilities** and **operational efficiency**.

Three types of processors organization

1. General Register Organization:

- It involves using multiple general-purpose registers for data storage during instruction execution.
- It allows direct access to operands and results, reducing memory access frequency.
- This organization enhances CPU speed and flexibility in executing operations.

2. Stack organization

- Stack Organization uses a stack to store and retrieve data.

- Instructions work by automatically using the top items on the stack, so no need to specify operands.
- This makes it simple but limits control over data access.

3. Accumulator-Based Organization :

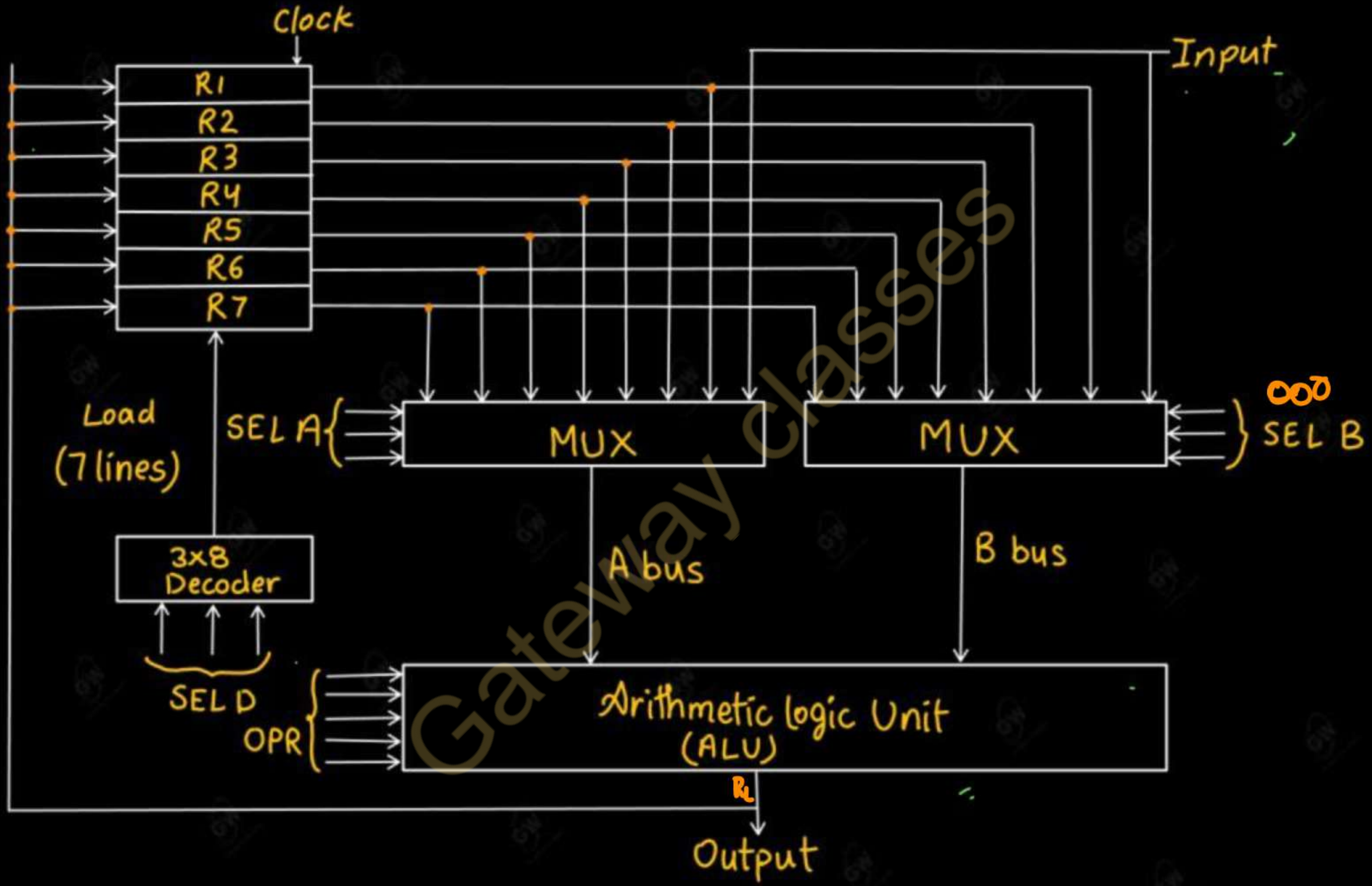
- It uses a single register, called the accumulator, to hold one operand and store results. All operations use the accumulator, making it simple but less flexible compared to multiple-register systems.

General Register organization

AKTU (2021-22/22-23)

- Generally CPU has **seven** general registers.
- Register organization show **how registers are selected** and **how data flow** between register and ALU.
- A **decoder** is used to select a particular register.
- The output of each register is connected to two multiplexers to form the two buses A and B.
- The **selection lines** in each multiplexer select the **input data for the particular bus**.

- The A and B buses form the two inputs of an ALU.
- The **operation select lines** decide the **micro operation** to be performed by ALU.
- The result of the micro operation is available at the output bus.
- The output bus connected to the inputs of all registers, thus by selecting a destination register it is possible to store the result in it.



Register and multiplexer input selection code

Binary code	SELA	SELB	SELD or SELREG
000	Input	Input	---
001	R1	R1	R1
010	R2	R2	R2
011	R3	R3	R3
100	R4	R4	R4
101	R5	R5	R5
110	R6	R6	R6
111	R7	R7	R7

Operation with symbol

OPR Select	Operation	Symbol
00000	Transfer A	TSFA
00001	Increment A	INCA
00010	Add $A + B$	ADD
00101	Subtract $A - B$	SUB
00110	Decrement A	DECA
01000	AND A and B	AND
01010	OR A and B	OR
01100	XOR A and B	XOR
01110	Complement A	COMA
10000	Shift right A	SHRA
11000	Shift left A	SHLA

EXAMPLE 1:

Micro-operation	SEL A	SELB	SELD	OPR
$R1 \leftarrow R2 - R3$	R2	R3	R1	SUB
CONTROL WORD	010	011	001	00101

EXAMPLE 2:

Micro-operation	SELA	SELB	SELD	OPR
$R7 \leftarrow R1$	R1	-	R7	TSFA
CONTROL WORD	001	000	111	00000

CONTROL WORD (2 Marks)

The combined value of a binary selection inputs specifies the control word.

SEL A (3 BITS)	SELB (3BITS)	SELREG OR SELD (3 BITS)	SELOPR (5 BITS)
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Micro-operation	SELA	SELB	SELD	OPR
Output $\leftarrow$ R2	R2	-	None	TSFA
CONTROL WORD	010	000	000	00000

✓ Advantages –

- ❑ Registers are quicker to access than memory, improving instruction execution speed.
- ❑ Multiple registers allow simultaneous operations, reducing the need to access memory frequently.
- ❑ General-purpose registers can hold data, addresses, or intermediate results, offering greater flexibility.

✓ Disadvantages –

- ❑ More registers require complex control logic and larger instruction sets, which can complicate CPU design
- ❑ Additional registers increase the power usage of the processor.
- ❑ Despite multiple registers, the space is still limited, which may require careful management in complex programs.

The function of control unit in general register organization .

- The control unit (CU) in a general register organization is responsible for directing the operations of the CPU by controlling the flow of data between the registers, ALU (Arithmetic Logic Unit), and memory.
- It generates the necessary control signals based on the instructions fetched from memory.

1. Instruction Fetching:

- The CU fetches the instruction from memory by generating the appropriate control signals
- This instruction is then placed into the instruction register (IR).
- The control unit uses the program counter (PC) to keep track of the next instruction's address.
- It updates the PC after fetching each instruction.

2. Instruction Decoding:

- Once the instruction is fetched, the CU **decodes it** to determine the **operation to be performed**.
- The decoding process involves breaking down the instruction into its opcode (operation code) and operands (data or addresses).

3. Control Signal Generation:

- Based on the **decoded instruction**, the CU generates **control signals that direct various components of the CPU on what actions to take**.
- These signals control data transfer between registers, instruct the ALU to perform arithmetic or logical operations, and manage data exchange between the CPU and memory.

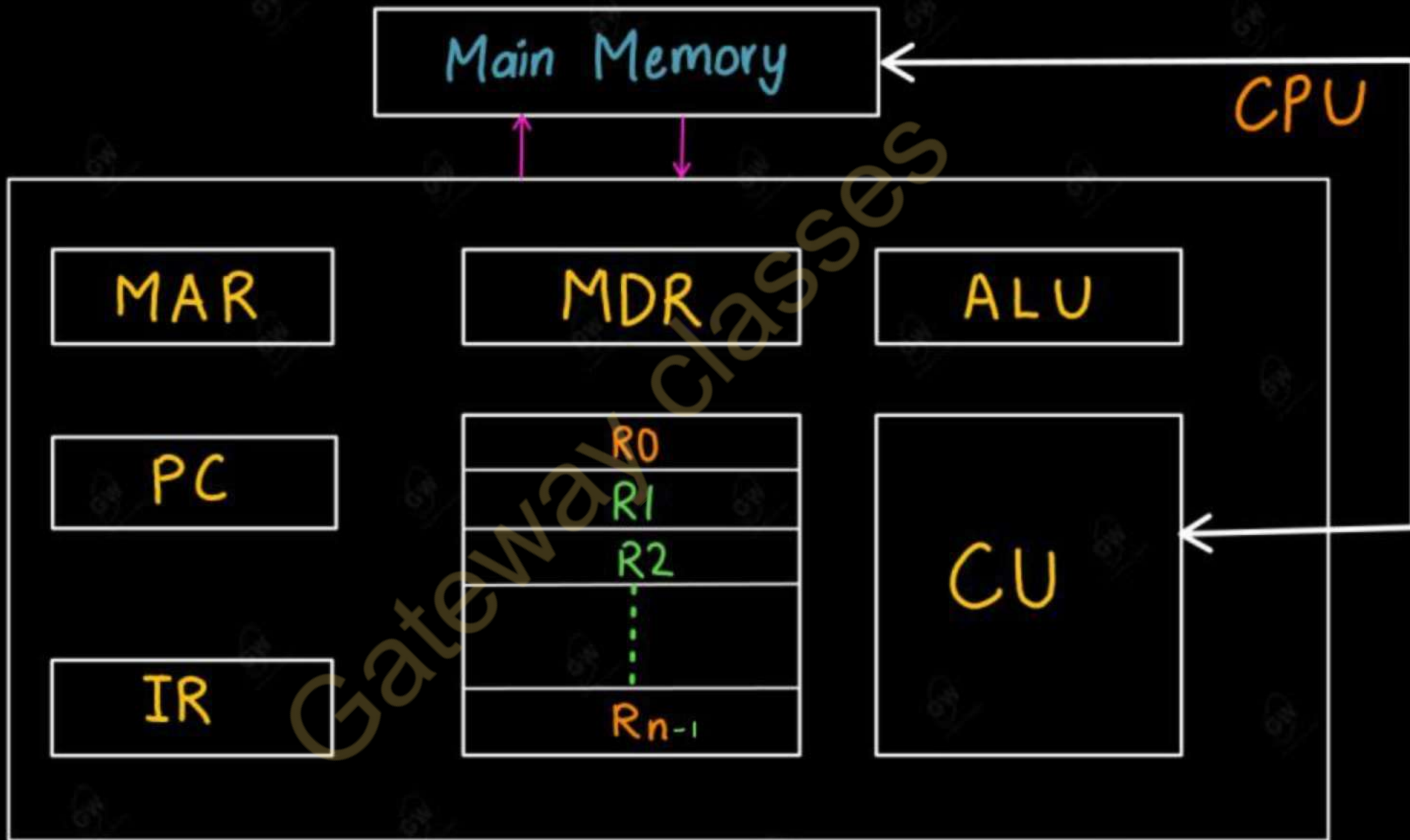
4. Execution Control:

- The CU controls the **execution of the instruction by coordinating** the sequence of operations needed to execute the instruction.
- **For example**, if the instruction is an addition operation, the CU will signal the ALU to perform the addition on the specified registers.

5. Memory and I/O Control:

- The CU manages the transfer of data between the CPU and memory, as well as input/output devices.
- It ensures that data is correctly read from or written to memory, and it controls the flow of data to and from peripheral devices.

You can also write about all Register



AKTU PYQS

Q.1	Discuss register organisation if the computer system has 7 register R1 TO R7 . How the operation <u>$R1 \leftarrow R2 + R3$</u> can be performed discuss,	AKTU 2021-22 AKTU 2014-15
Q.2 ✓	Explain the function of control unit in register organisation with the help of diagram $(CPU + M)$	AKTU 2023-24
Q.3 ✓	Explain the general purpose register based organisation	AKTU 2021-22
Q.4 ✓	What do you mean by processor organisation? Explain various types of <u>processor organisation</u>	AKTU 2023-24 AKTU 2020-21 AKTU 2014-15

Computer organization and architecture

UNIT 1 INTRODUCTION

Today's Target

- Accumulator based Organization
- Stack Organization
- AKTU PYQs

By PRAGYA RAJVANSHI

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Accumulator-Based Organization

- It is sometimes called single accumulator – based organization.
- In this type of CPU organization, the accumulator register is used implicitly for processing all instructions of a program and storing the results into the accumulator.
- The instruction format that is used by this CPU Organization is the One address field. Due to this, the CPU is known as One Address Machine.

The main points about Single Accumulator based CPU Organization are:

- In this CPU Organization, the first ALU operand is always stored into the Accumulator and the second operand is present either in Registers or in the Memory. $2+3$
- Accumulator is the default address thus after data manipulation the results are stored into the accumulator.
- One address instruction is used in this type of organization.

Mainly two types of operation are performed in a single accumulator based CPU organization:

1. Data transfer operation

For example: LOAD X, STORE Y

- **LOAD** is a memory read operation that is data is transferred from memory to accumulator
- **STORE** is a memory write operation that is data is transferred from the accumulator to memory

2. ALU operation

Arithmetic operations are performed on the data.

For example: MUL X

$$\underline{AC \leftarrow AC * M[X]}$$

AC is the Accumulator and M[X] is the memory word located at location X.

$$x = (a * b) + (c * d)$$

Load a
 MUL b
 STORE y
 LOAD c
 MUL d
 ADD y
 STORE x

$AC \leftarrow M[a]$
 $AC \leftarrow AC * M[b]$
 $M[y] \leftarrow AC$
 $AC \leftarrow M[c]$
 $AC \leftarrow AC * M[d]$
 $AC \leftarrow AC + M[y]$
 $M[x] \leftarrow AC$

01011	1	a
0111	2	b
	3	c
	4	d
	2	y
	14	x

3
 AC

12 14

✓ Advantages –

- One of the operands is always held by the **accumulator register**. This results in **short instructions and less memory space**.
- The **instruction cycle takes less time** because it saves time in instruction fetching from memory.

✓ Disadvantages –

- When **complex expressions** are computed, **program size increases** due to the usage of many short instructions to execute it. Thus memory size increases.
- As the number of instructions increases for a program, the **execution time increases**.

Stack Organization(AKTU 2022-23)

- Stack organization is a method of managing data in a computer system using a stack data structure.
- A stack is a collection of elements with two primary operations: push (to add an item to the stack) and pop (to remove the item from the top of the stack).
- It follows the Last In, First Out (LIFO) principle, meaning the most recently added item is the first to be removed.

The stack can be implemented using two ways

- Register Stack .
- Memory Stack.

Implementation of register as a stack/

Register stack organisation

- 64 general registers are used for stack implementation, so 6 bits are required to address 64 registers.
- SP is a stack pointer (top of the stack) that stores the location's address where we must push or pop any data. As SP stores the address of that location where we PUSH or POP data, its size will be 6 bits.
- We use two flag registers named FULL and EMPTY.

- If the value of FULL is 1, then Empty will be zero, and vice versa.
$$\begin{array}{r} 1 - 10 \\ 0 - 9 \end{array}$$
- If the FULL value is 1, then there is no more space to enter any new data.
- If the EMPTY value is 1, then it means we can add new data in stack-organized registers.

$$\begin{array}{r} 2 \\ + 3 \\ \hline 5 \end{array} \quad \begin{array}{r} 1 \\ 0 \\ 0 \\ \hline 1 \end{array}$$

$$2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$$

$$\begin{array}{r} 2^3 \ 2^2 \ 2^1 \ 2^0 \\ 8 \ 4 \ 2 \ 1 \\ 0 \ 0 \ 1 \ 0 \quad 2 \\ 0 \ 0 \ 1 \ 1 \quad 3 \\ \hline 0 \ 1 \ 0 \ 1 \quad 5 \end{array}$$

$$\begin{array}{l} 0+1=1 \\ 1+0=1 \\ 0+0=0 \\ 1+1=1 \text{ with carry } 1 \end{array}$$

$$\begin{array}{r} 8 \\ 2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0 \\ 16 \ 8 \ 4 \ 2 \ 1 \\ 0 \ 1 \ 0 \ 0 \ 0 \quad -8 \\ 1 \ 0 \ 1 \ 1 \ 1 \\ + 1 \\ \hline 1 \ 1 \ 0 \ 0 \ 0 \\ +ve \quad \boxed{0 \ 1 \ 0 \ 0 \ 0} \quad -8 \\ -ve \quad \boxed{1 \ 1 \ 0 \ 0 \ 0} \rightarrow -8 \\ -16+8=8 \end{array}$$

1's complement
2's complement

$$\begin{array}{r} 2^3 \ 2^2 \ 2^1 \ 2^0 \quad -12 \\ 16 \ 8 \ 4 \ 2 \ 1 \\ 0 \ 1 \ 1 \ 0 \ 0 \quad -12 \\ 1 \ 0 \ 0 \ 1 \ 1 \\ + 1 \\ \hline 1 \ 0 \ 1 \ 0 \ 0 \end{array}$$

$$-16+4=12$$

Register Stack Organization

Full

Empty

PUSH

$SP \leftarrow SP + 1$

$M[SP] \leftarrow DR$

if $(SP = 0)$ then

$(FULL \leftarrow 1) \text{ AND } (EMPTY \leftarrow 0)$

POP

$DR \leftarrow M[SP]$

$SP \leftarrow SP - 1$

if $(SP = 0)$ then

$(EMPTY \leftarrow 1) \text{ AND } (FULL \leftarrow 0)$

SP



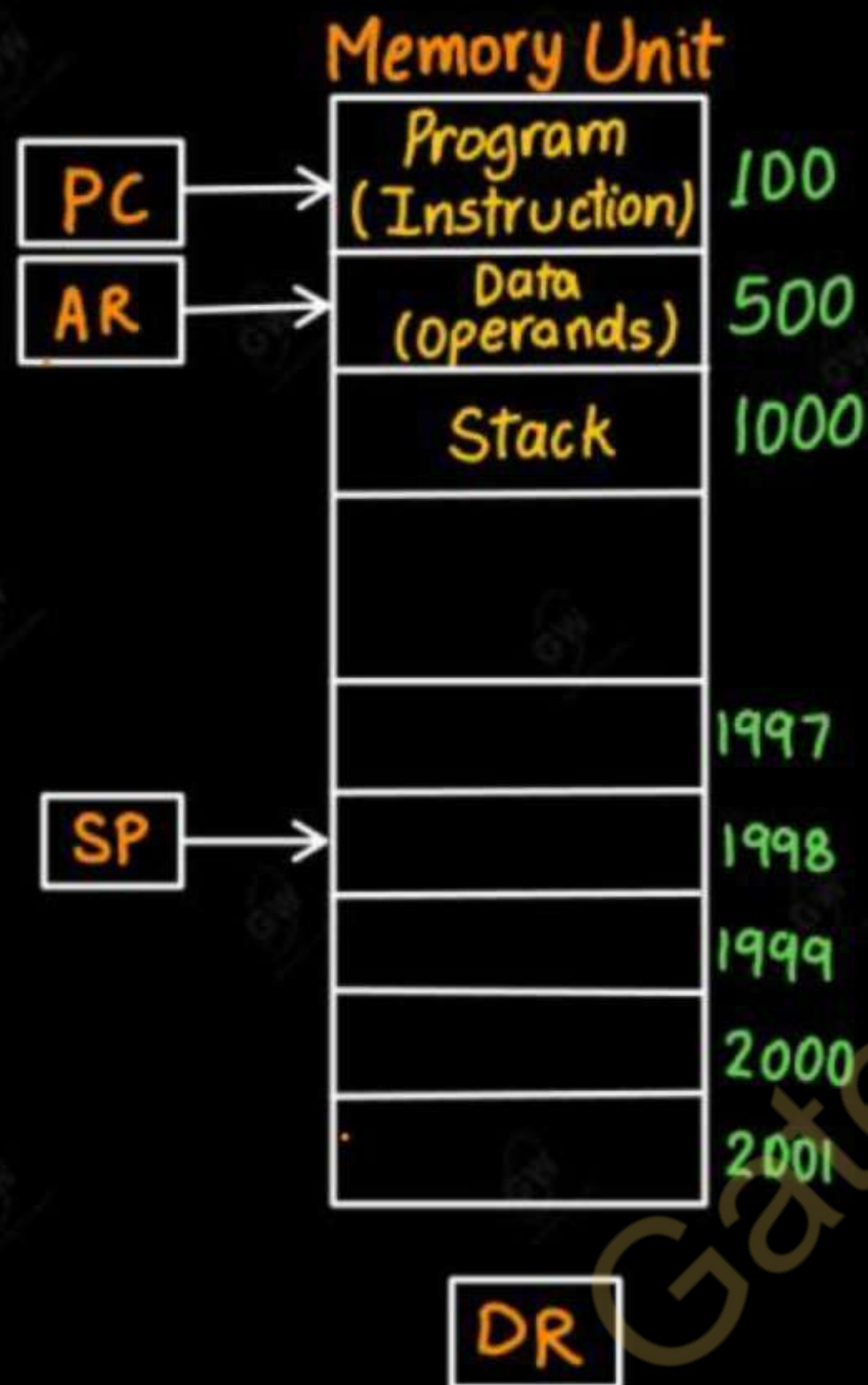
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Implementation of memory as a stack/

Memory Stack Organization

- A portion of memory is assigned to a stack operation to implement the stack in the CPU.
- The portion of computer memory divided into three segments: Program Instructions, Data, and Stack.
- Program Counter (PC): It is a register that points to the address of the next instruction that is going to be executed in the program.
- Address Register (AR): This register points at the collection of data and is used during the execute phase to read an operand.

Stack Pointer (SP): It points at the top of the stack and is used to push or pop the data items in or from the stack



➤ SP points to a beginning value '2001'. Therefore, the stack increase with decreasing addresses. The first element is saved at address 2000, the next element is saved at address 1999 and the last element is saved at address 1000.

PUSH OPERATION	POP OPERATION
$SP \leftarrow SP - 1$ $K[SP] \leftarrow DR$	$DR \leftarrow K[SP]$ $SP \leftarrow SP + 1$

AKTU PYQS

Q1	What is stack organisation ? Compare register stack and memory stack and register stack	(AKTU 2021-22)
Q2	Discuss memory stack and register stack	(AKTU 2018-19) (AKTU 2019-20)

UNIT 1 INTRODUCTION

Today's Target

- Difference between Memory and Register stack
- AKTU PYQs

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Difference between -(AKTU 2023-24/2021-22)

Features	Memory Stack	Register Stack
Location	Located in system RAM.	Located in CPU registers.
Size	Generally larger, limited by RAM.	Smaller, limited by number of registers.
Access Time	Slower access due to memory latency.	Faster access due to proximity to CPU.
Usage	Stores function call information, local variables.	Stores temporary data during computation.
Management	Managed by OS and compiler.	Managed directly by CPU.
Overflow	Can lead to stack overflow if exceeded.	Overflow is typically handled by CPU or results in errors.
Access Method	Accessed via memory addresses.	Accessed via CPU registers directly.

Difference between-

Features	Memory Stack	Register Stack
Operations	Push and pop operations; LIFO (Last In, First Out).	Push and pop operations; LIFO (Last In, First Out).
Performance Impact	<u>Slower</u> due to memory access delays.	<u>Faster</u> , as <u>registers</u> are part of the CPU.
<u>Initialization</u>	Initialized by the <u>operating system</u> and <u>compiler</u> .	<u>Initialized by the CPU automatically during execution.</u>
Scalability	<u>Can grow large as long as RAM allows.</u>	<u>Limited by number of hardware register.</u>
Functionality	Used for managing function calls and local variables.	Limited by the number of available CPU Used for fast access to frequently used data registers.

301. The address field has the value 400 A processor register(r1) contains the value 200 and Evaluate the effective address if the addressing mode of the instruction is

(AKTU 2018-19)

(a) Direct

(b) Indirect

(c) Immediate

(d) Register Indirect /direct

(e) Relative and indexed (consider r1 as index register)

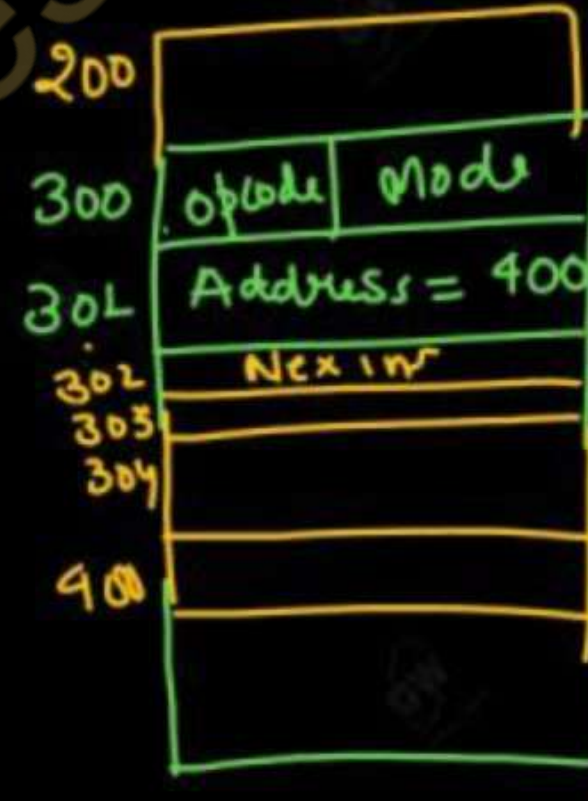
(f) Auto increment

(g) Auto decrement



PC = 302

$$\begin{aligned} \text{Relative} &= \text{PC} + \text{Address} \\ &= 302 + 400 \\ &= 702 \end{aligned}$$



r1 = 200

Indirect Register mode

X

X

ADDRESSING MODE	EFFECTIVE ADDRESS
Implied/Implicit	—
Stack	—
Immediate	301
Direct	400
Indirect	$M[400]$
Register direct	R_1 (Register name)
Register indirect	200
Relative	$PC + Address = 302 + 400 = 702$
Indexed	$Indexed\ Register + Address\ part = 200 + 400 = 600$
Auto-increment	200
Auto-Decrement	$200 - 1 = 199$

ADDRESS		
200	LOAD TO AC	MODE
201	ADDRESS=500	
202	NEXT INSTRUCTION	

399	450	
400	700	
500	800	
600	900	
700	333	
702	325	
800	300	

Q2 (2 Marks)

Q2 The effective address is the actual memory location where data is accessed or stored, computed by using the addressing mode of an instruction

PC=200 , 202 (After Executing the Current Instⁿ)

R1=400

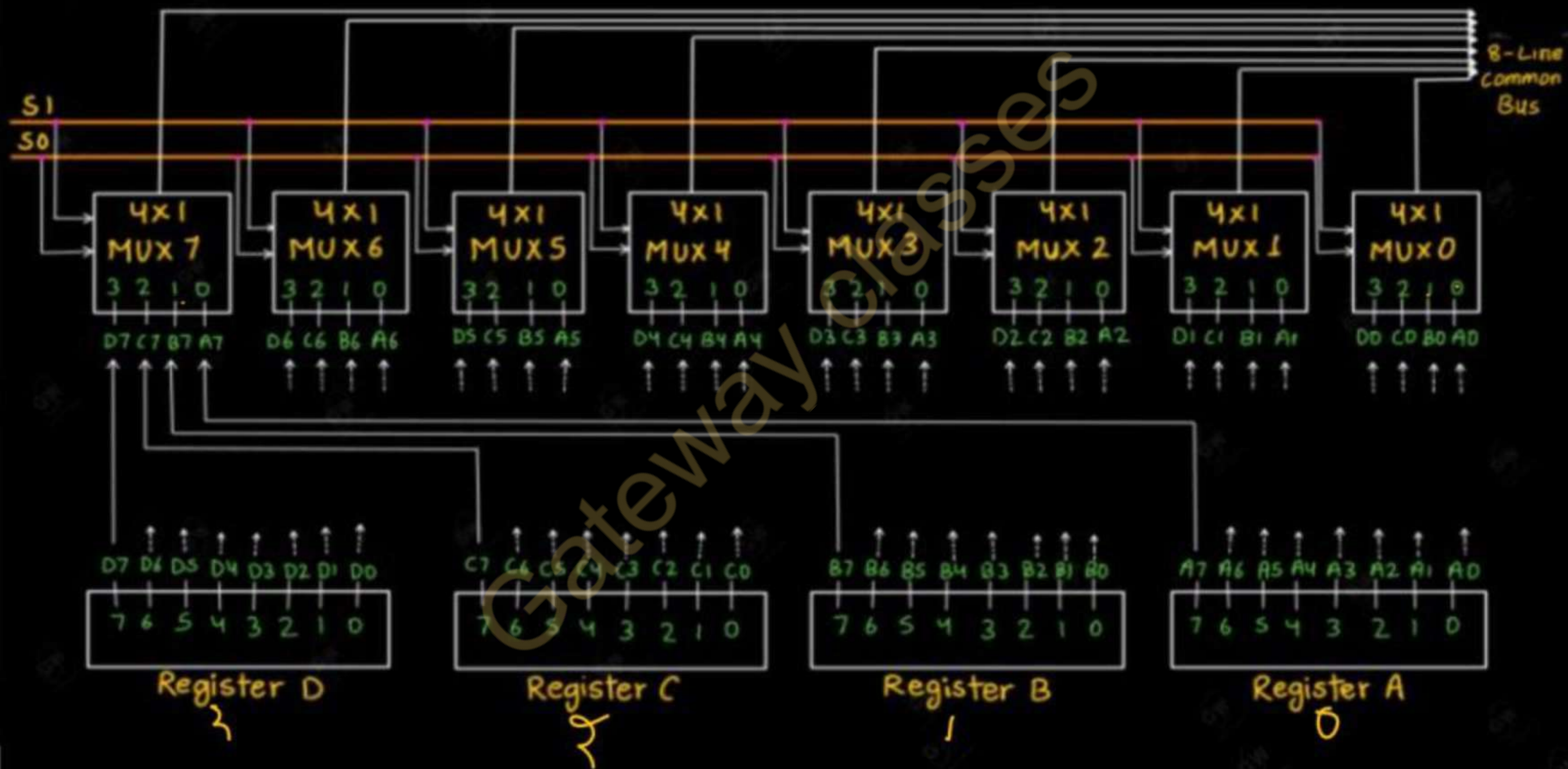
XR=100(INDEXED REGISTER)

AC



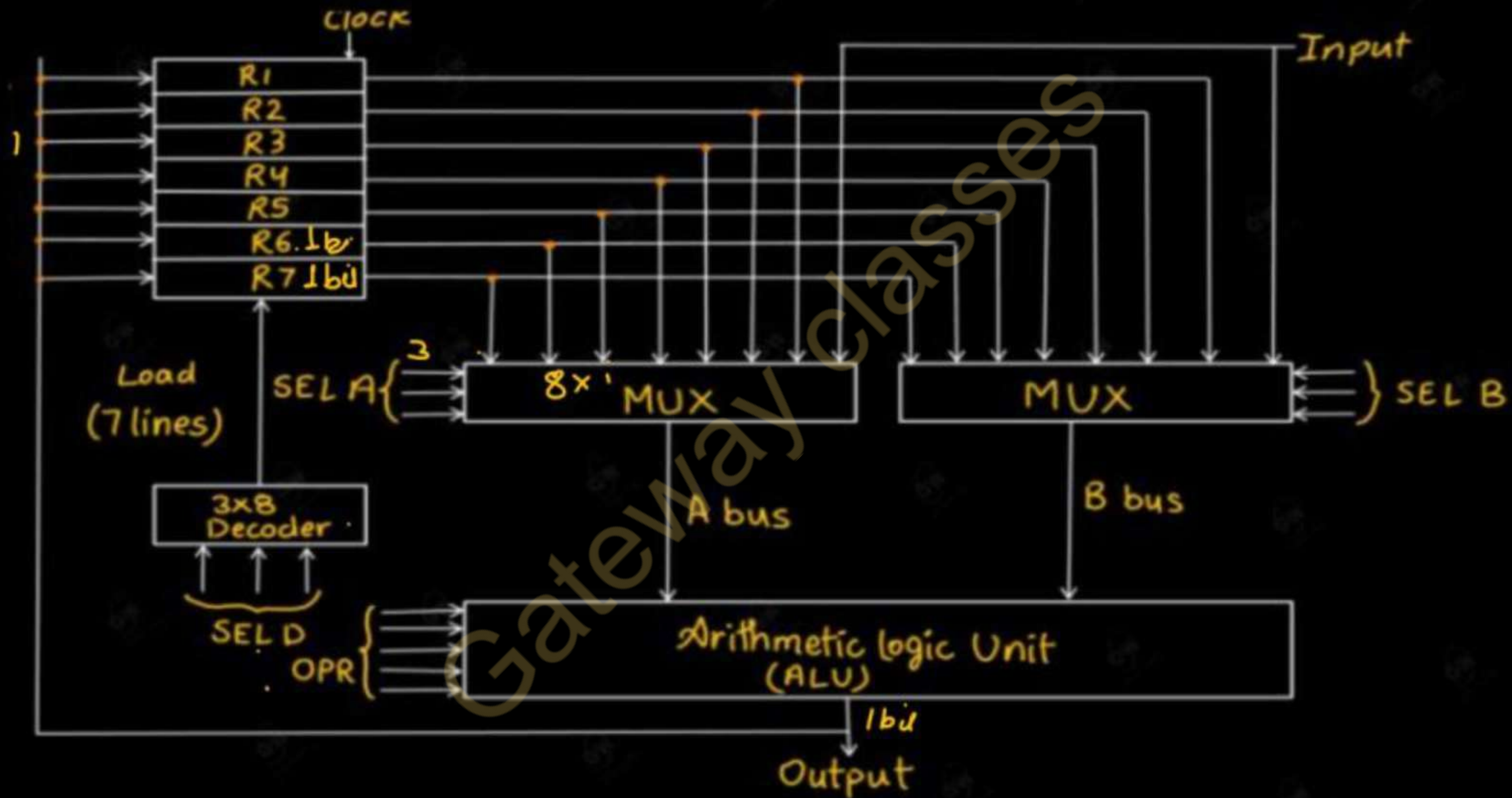
ADDRESSING MODE	EFFECTIVE ADDRESS	CONTENT OF AC/CONTENT OF EFFECTIVE ADDRESS
Immediate	201	500
Direct	500	800
Indirect	800	300
Register direct	R1	400
Register indirect	400	700
Relative	$PC + Address = 202 + 500 = 702$	325
Indexed	$XR + Address = 100 + 500 = 600$	900
Auto-increment	400	700
Auto-Decrement	399	450

COMMON BUS TRANSFER



- ① Number of Multiplexer = Number of bits in a register
 - ② Number of input line = Number of Register
(n)
 - ③ No of Selection line = $\log_2 n$
- Size of Multiplexer = $n \times 1$

General register organization



Question 3 A bus-organized CPU has 16 registers with 32 bits in each, an ALU, and a destination decoder.

1. How many multiplexer are there in A bus what is the size of multiplexers?
 2. How many selection lines are needed for multiplexers (MUX A AND MUX B)
 3. How many input(select) and output line in decoder.
 4. How many input and output are there in the ALU for data including input and output carrier
3. Formulate a control word for the system assuming that ALU has 35 operations.

(AKTU 2019-20)

(Bus A | Bus B)
 No of Multiplexer = No of bits in a Register = 32
 Number of input lines = Number of Register = 16

Size of Multiplexer = No of input lines $\times$ 1
 16 $\times$ 1

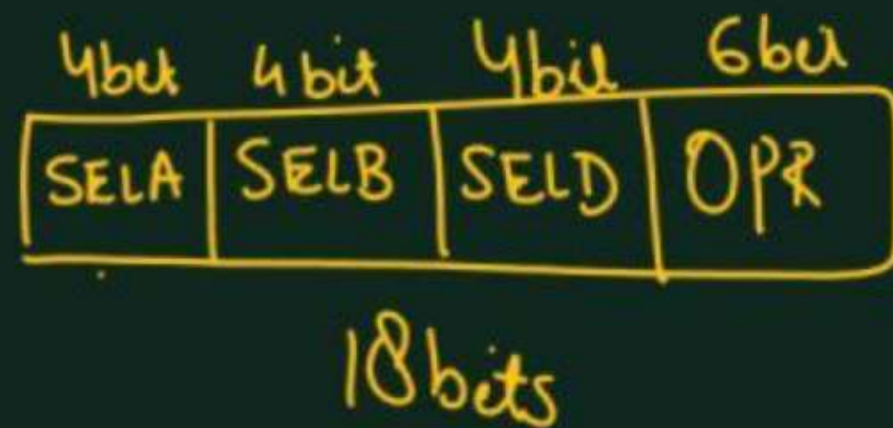
No of Selection line = 4
 Bus A Bus B
 (MUX)

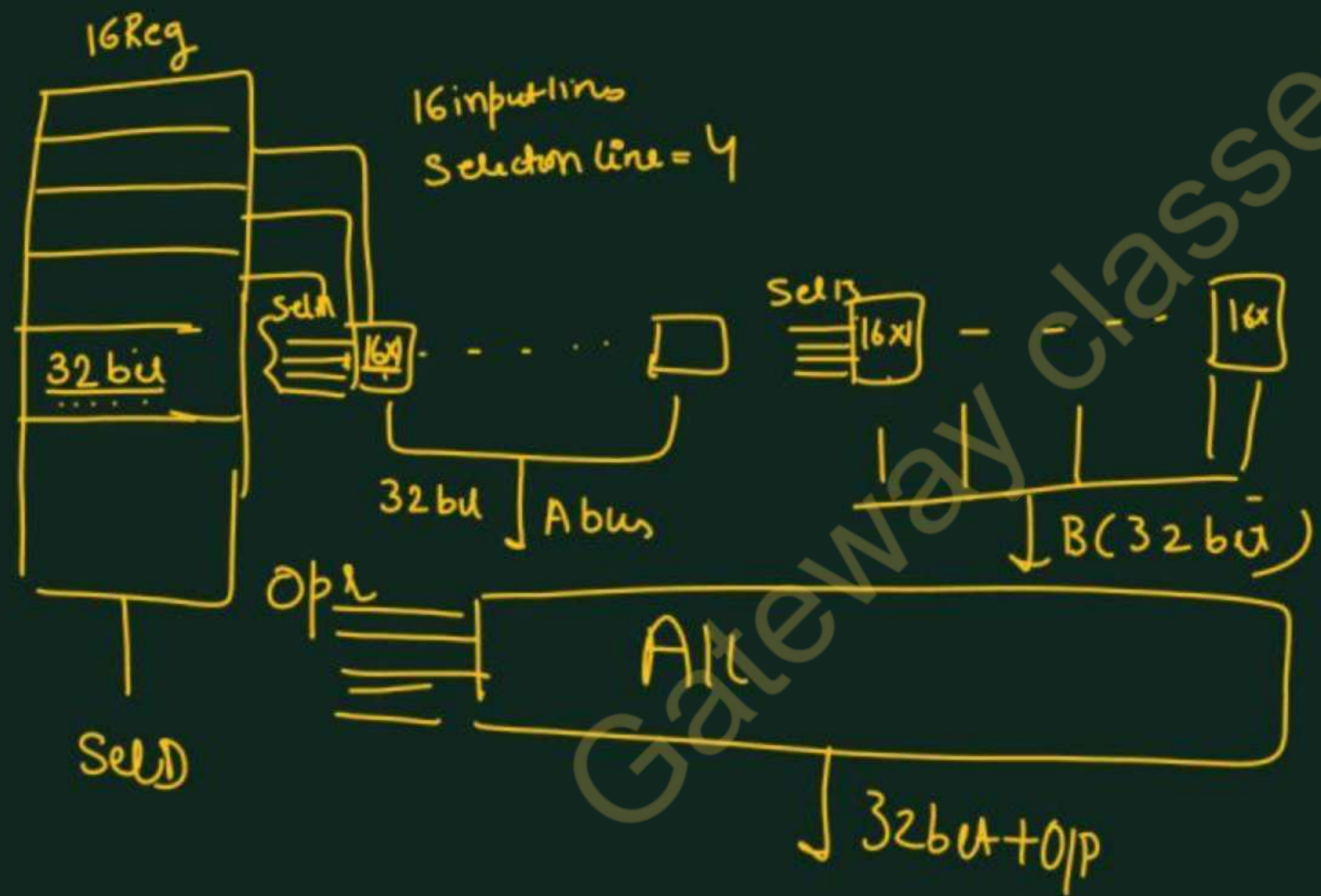
Input lines in decoder = 4 (Selection line)
 Output line = 16
 of
 Decoder

$$\begin{aligned} 2^0 &= 1 \\ 2^1 &= 2 \\ 2^2 &= 4 \\ 2^3 &= 8 \\ 2^4 &= 16 \end{aligned}$$

Input lines in ALU = 32 bit + 32 bit + 1 bit
 (from Bus A) (from Bus B) (Input carry)
 65 bits

Output lines in ALU = 32 bit + Output (1 bit)
 Carry
 33 bit





Question 4 A bus-organized CPU has 8 registers with 16 bits in each, an ALU, and a destination decoder.

1. How many multiplexers are there in a bus and what is the size of multiplexers?
2. How many selection lines are needed for multiplexers and decoder?
3. Formulate a control word for the system assuming that ALU has 35 operations.

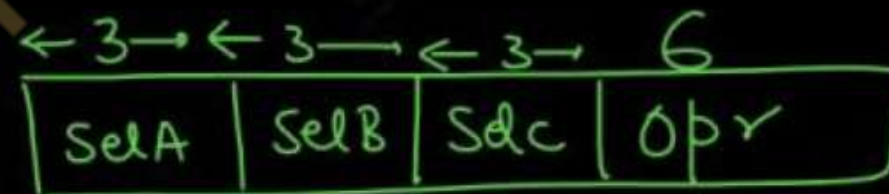
(AKTU2018-19)

1. No of Multiplexer = No of bits in a Register = 16
Bus A

No of input line = No of Register = 8

Size of Multiplexer = Number of input line $\times$ 1
8 $\times$ 1

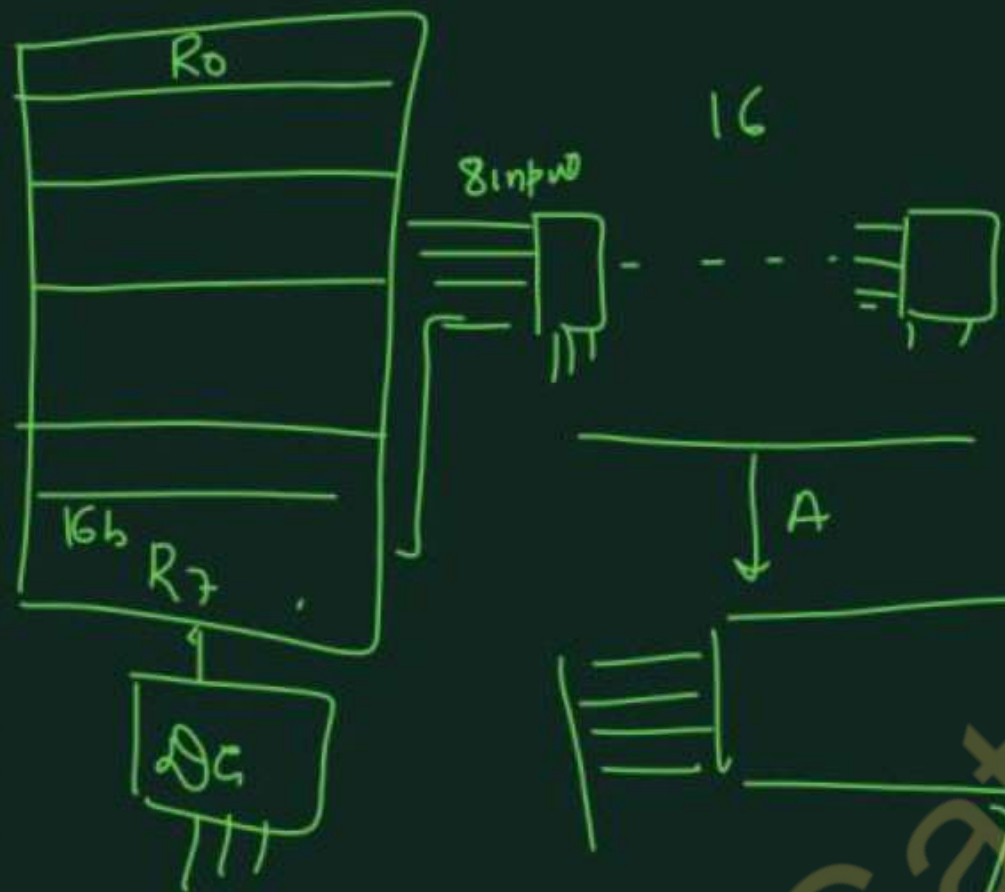
2. Number of Selection = 3
in Multiplexer & decoder



15 bits

2^0	→	1
2^1	→	2
2^2	→	4
2^3	→	8
2^4	→	16
2^5	→	32
2^6	→	64

$2^0 \rightarrow 1$
 $2^1 \rightarrow 2$
 $2^2 \rightarrow 4$
 $2^3 \rightarrow 8$



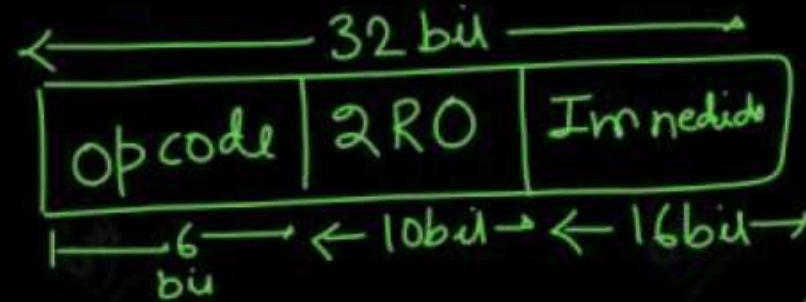
$11 \rightarrow 3$
 $10 \rightarrow 2$
 $01 \rightarrow 1$
 $00 \rightarrow 0$

65
 1111
 $+1111$
 $\hline 11110$
 64
 $+2$
 $\hline 128$

$2^0 \rightarrow 1$
 $2^1 \rightarrow 2$
 $2^2 \rightarrow 4$
 $2^3 \rightarrow 8$
 $2^4 \rightarrow 16$
 $2^5 \rightarrow 32$
 $2^6 \rightarrow 64$
 $2^7 \rightarrow 128$

32 bit instruction word has an opcode, two registers operands and an immediate operand.

The number of bits available for the immediate operand field is _____. (AKTU 2020-21)



Number of Distinct Insⁿ = 40
 Number of bit to Represent 1 Instruction = 6
 Number of Register = 24
 Number of bit to Represent 1 register = 5
 2 Register operand = 10 bit
 Immediate operand bit = Total instruction size - (10 + 6)
 32 - 16 = 16 bit

$$\begin{array}{r} 32 \\ 16 \\ \hline 16 \end{array}$$

$$\begin{array}{r} 256 \\ \times 2 \\ \hline 512 \end{array}$$

2 ⁰	1
2 ¹	2
2 ²	4
2 ³	8
2 ⁴	16
2 ⁵	32
2 ⁶	64
2 ⁷	128
2 ⁸	256
2 ⁹	512
2 ¹⁰	1024

UNIT 1 INTRODUCTION

Today's Target

- Bus architecture
- Multiple bus hierarchies
- AKTU PYQs

By PRAGYA RAJVANSHI
B.Tech, M.Tech(C.S.E.)

Q.1 The instruction is stored at location 700 with its address fields at location 701. The address field has the value 1000. A processor register contains the value 500 and index register contains the value 300. Evaluate the effective address if the addressing mode of the instruction is (a) Direct (b) Indirect (c) Immediate (d) Relative (e) Register Indirect (f) Auto increment (g) Auto decrement **(AKTU 2023-24)**.

700	Opcode Mode
701	Address = 1000
702	Next
703	
704	

Let process register

$R_1 = 500$

$IX = 300$

(Index Register)

$PC = 702$

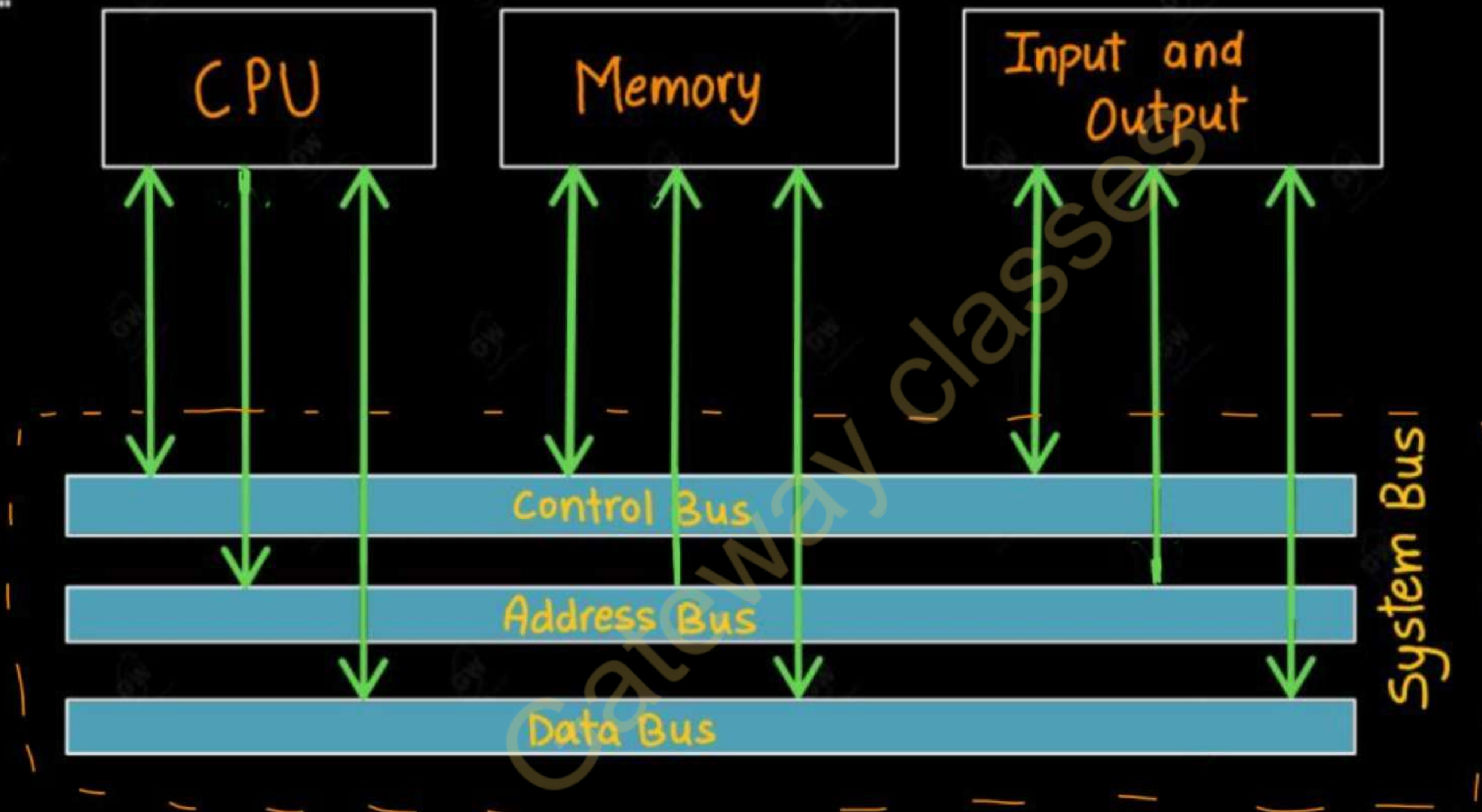
Direct $\rightarrow$ EA 1000
 Indirect $\rightarrow$ M[1000]
 Immediate $\rightarrow$ 701
 Relative = $PC + \text{Address}$
 $702 + 1000$
 $= 1702$
 Register indirect $\rightarrow$ 500
 Auto increment $\rightarrow$ 500
 Auto decrement $\rightarrow$ $500 - 1$
 499

Mode	Op Code	Address
------	---------	---------

If a two word instruction is start at a memory location 200 and its address 500 is stored at location 201. If the contents at location 400 is 700, 500 is 800, 600 is 900, 702 is 325, and 800 is 300 and a processor register R has the value 400 Evaluate the effective address and the contents of AC if the addressing modes are immediate, direct, indirect, register, register indirect, relative and indexed. (AKTU 2018-19)

Memory Location		Index Reg	Effective Address	AC Content
200	Op code Mode	R=400	201	500
201	Address = 500	PC=202	500	800
400	700		800	300
500	800		R	400
600	900		400	700
702	325		PC + Address	325
800	300		$202 + 500 = 702$	M[900]
			$400 + 500 = 900$	

Bus architecture (AKTU 2018-19)



A+B
Infix

AB+
+AB

postfix ✓
prefix

2*3+5

2*(3-5)
(*(-

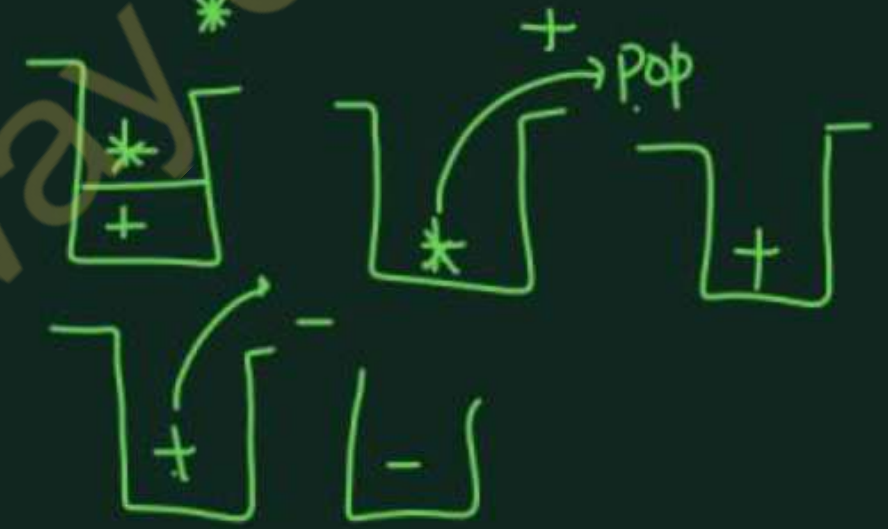
A, B are operand
+ operator

A+B*C
A+BC*
A+x
Ax+
ABC*+

(*(*

Operator-precedence

$\frac{\wedge}{* /} \frac{+ -}{(* -)}$



2*3|5
6|5

2+3-5
5-5=

Associates
left to Right

$$(A + B / C * (D + E) - F)$$

$$A + B / C * (D + E) - F$$

$$A + B / C * D + E - F$$

$$A + B / C * D + E - F$$

$$A + X * Y - F$$

$$A + B / C * D + E - F$$

$$A + Z - F$$

$$A + B / C * D + E - F$$

$$A + B / C * D + E - F$$

	Symbol	Stack	Postfix
	(	(	-
	A	(	A
	+	(	A
	B	(+	AB
	/	(+	AB
	C	(+ /	AB C
	*	(+ /	AB C
	(	(+ *	AB C /
	D	(+ * (	AB C /
	+	(+ * (+	AB C / D
	E	(+ * (+	AB C / D E
	)	(+ *	AB C / D E +
	-	(-	AB C / D E + *
	F	(-	AB C / D E + * +
	)		AB C / D E + * + F
			AB C / D E + * + F -



How operation performed by the stack evaluate (AKTU 2022-23)

X: $5+6*(8-2*3)$

$$(5+6*(8-2*3)) \quad 5+6*2 = \underline{17}$$

How expression arte evaluated using stack (AKTU 2018-19)

Symbol	Stack	Postfix
(	(	—
5	(	5
+	(+	5
6	(+	5 6
*	(+ *	5 6
(	(+ * (	5 6
8	(+ * (	5 6 8

—	(+ * (-	5 6 8
2	(+ * (-	5 6 8 2
*	(+ * (- *	5 6 8 2
3	(+ * (- *	5 6 8 2 3
)	(+ *	5 6 8 2 3 * -
)	<u>—</u>	5 6 8 2 3 * - * +

56823*-*+

Symbol
Scanned.

5
6
8
2
3
*
-
*
+

Stack

5
56
568
5682
56823
5686
562
512
17

(Short trick)

$$5 + 6 * (8 - 2 * 3)$$

$$5 + 6 * (8 - 2 * 3)$$

$$5 + 6 * 8 - 2 * 3$$

$$5 + 6 * 8 - 2 * 3$$

$$5 + 6 * 8 - 2 * 3$$

MULTIPLE BUS HEIRARCHY

- A multiple bus hierarchy in computer architecture refers to a system design where multiple buses are used to connect different components, such as CPUs, memory, and I/O devices.

Need for Multiple Bus Hierarchy in Computer Architecture:

1. Performance Improvement:

- A single bus architecture often becomes a **bottleneck** as it forces all data transfers through one bus, leading to congestion.
- Multiple bus hierarchy allows different components (CPU, memory, I/O devices) to communicate simultaneously, reducing wait times and improving overall system performance.

3. Parallelism:

- With multiple buses, data transfers can occur in parallel across different buses, enabling multiple operations to be carried out simultaneously.

4. Scalability:

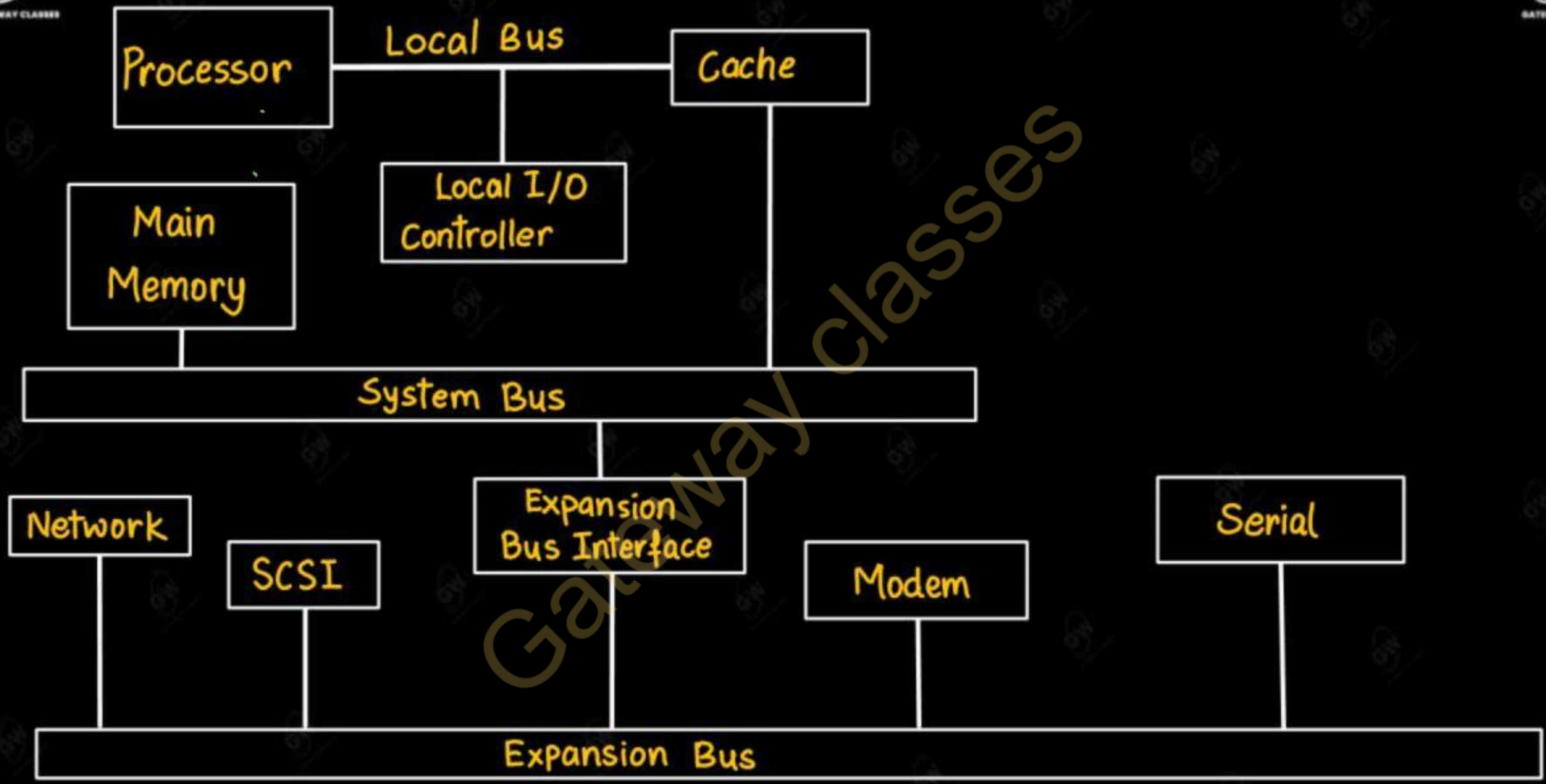
- As systems grow in complexity (e.g., adding more CPUs, memory modules, or I/O devices), a single bus can become overwhelmed.
- Multiple bus hierarchy provides scalability by distributing the load across several

buses, ensuring that the system can handle increased demands without significant performance degradation.

5. Reduced Latency:

- In a single bus system, components often have to wait for the bus to become available, increasing latency.
- A multiple bus hierarchy reduces this latency by allowing different buses to handle specific tasks (e.g., a dedicated memory bus), ensuring quicker data access and transfer.

Traditional Hierarchy Bus Architecture

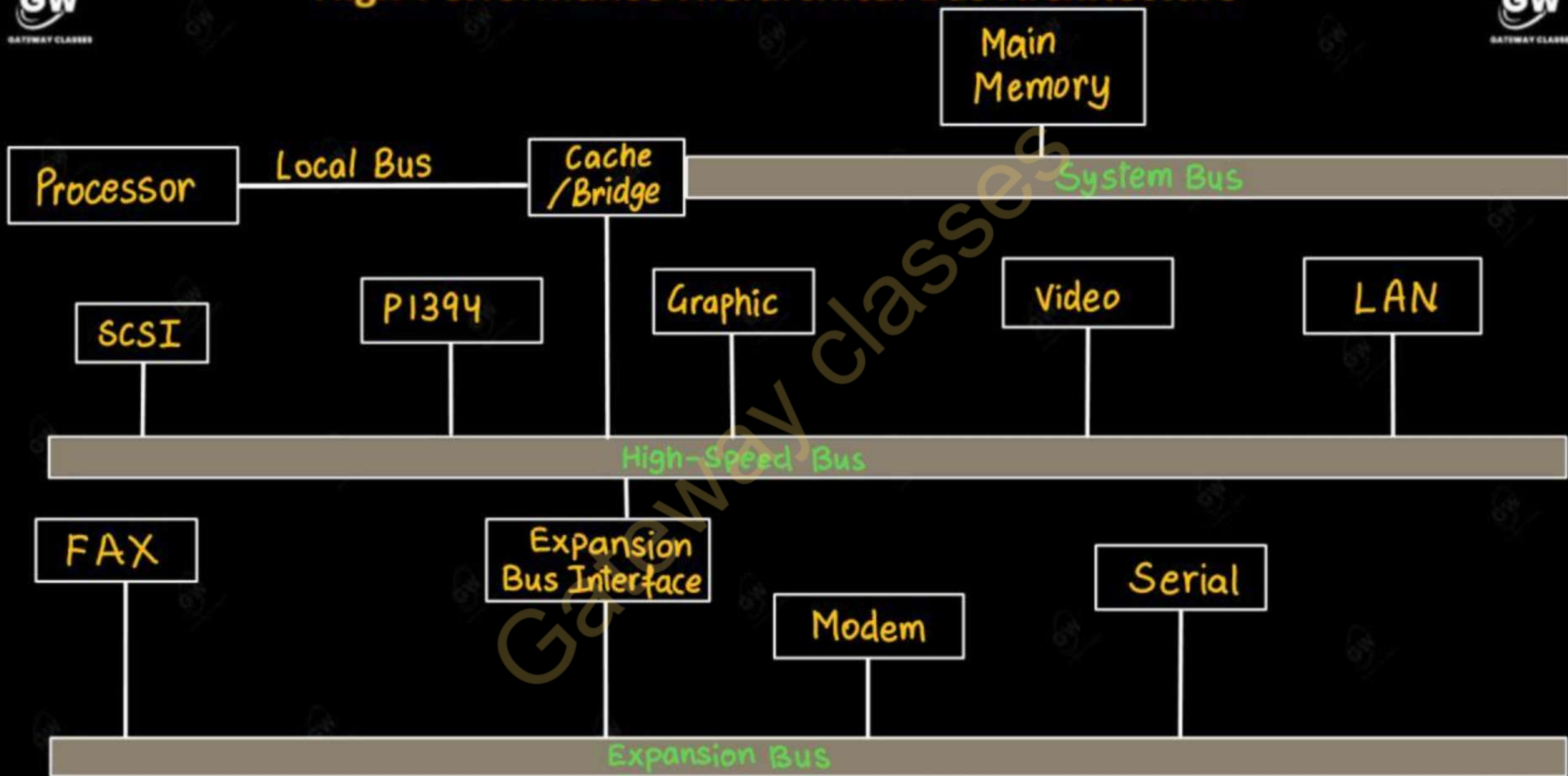


- Isolate processor to memory traffic from i/o traffic
- Cache memory act as an interface to system bus
- Expansion bus interface it to the external devices
- Local bus support cache memory and one or more local devices
- Cache memory attached o main memory
- Expansion bus interface buffer data transfer between system bus and i/o bus.

This arrangement allows the system to support a wide variety of I/O devices.

It This architecture is efficient.

High-Performance Hierarchical Bus Architecture



- This architecture requires a bridge between the processor's bus and the high speed bus.
- There is a local bus that connects the processor to a cache controller, which in turn is connected to a system bus that supports main memory
- The cache controller is integrated into a bridge or buffering device, that connects to the high-speed bus. This bus supports connection to high speed LANs

Thank
you